Paraconsistent Reactive Graphs: Their Semantics and Conflict Resolution Policies

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Abstract

This paper introduces *Paraconsistent Reactive Graphs*, an extension of Reactive Graphs that incorporates paraconsistency into the graph edges to naturally address the vagueness and inconsistency often encountered in dynamic systems. In these systems, the dynamic behavior is given by a special kind of edges, called *hyper-edges*, which act as switches that either activate or deactivate others. Furthermore, edges in Paraconsistent Reactive Graphs are not bivalent; instead, the framework allows for the representation of vague, consistent, and inconsistent information. This is potentially relevant when modeling inputs from multiple sources that may be individually consistent but collectively incomplete or conflicting.

Consequently, this work raises fundamental questions about conflict resolution, namely, how to manage and combine information even in the presence of contradictions. To address this, we propose and motivate, a few strategies for combining information from various sources. Finally, a semantics for Paraconsistent Reactive Graphs is presented and results under which consistency is preserved are explored.

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1 Introduction

Dov Gabbay introduced *reactive graphs* as transition structures whose accessibility relation can be dynamically modified by prior transitions [27, 29]. This dynamic behavior is specified by higher-order edges, or hyper-edges, which update the accessibility relation when traversed. Indeed hyper-edges connect one edge to another, acting as switches that either activate ($\twoheadrightarrow$) or deactivate ($\multimap$) the connected edge [26]. Thus, a reactive graph consists of ordinary transitions, called *ground edges*, enriched with hyper-edges that control activation of other edges. Reactive graphs have proven effective in modeling a wide range of reactive systems [27, 28, 41].

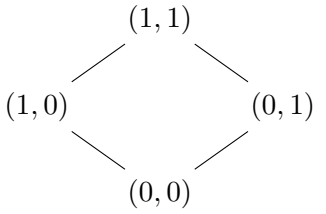
However, certain practical applications involve input from multiple, possibly conflicting sources. In such contexts, the classical bivalent structure of reactive graphs may be insufficient. This motivates the exploration of extensions that can handle vague and inconsistent information in a sound and controlled manner.

One such extension introduces fuzziness in both ground and hyper-edges which are assigned values in the real interval $[0, 1]$, capturing varying degrees of certainty [7, 8, 9, 41]. This framework allows modeling systems with vague behavior that can be explored using a fuzzy modal logic. Furthermore, this was applied to model scenarios in biology [41], for example, the circadian rhythm of cyanobacteria, a biological system represented by a regulatory network of components such as genes and proteins. In such networks, one component may induce (via activation) or inhibit (via deactivation) the production of another. Moreover, transitions in these systems may occur with varying degrees of likelihood, making fuzzy reactive graphs a natural and expressive framework for their specification.

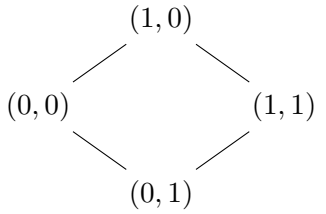
Building on this idea, we propose a paraconsistent extension of reactive graphs that accommodates both uncertainty and inconsistency. In this work, the term *paraconsistency* is used in a precise sense to indicate that the framework tolerates contradictions without leading to trivialization. Formally, paraconsistency is a property of a consequence relation: a logic is said to be paraconsistent if its consequence relation is *non-explosive*, meaning that from a contradiction $A, \neg A$, it is not possible to derive an arbitrary conclusion B. Thus, paraconsistent reasoning offers a sound and controlled way to reason in the presence of contradictions.

One of the founding contributions to the field of paraconsistency is Jaśkowski's *discursive logic*, introduced in 1948 [**?**]. Jaśkowski was inspired by group discussions in which participants may each provide self-consistent information, yet when combined may yield contradictions. Since Jaśkowski's work, extensive developments by Asenjo, Carnielli, da Costa, and others have established paraconsistent reasoning as a well-developed area of study [3, 10, 20], see also [25] for a comprehensive overview.

Our starting point is Belnap-Dunn four-valued lattice [5], where each truth value is a pair $(\#t, \#f) \in \{0,1\}^2$. Informally, as discussed in [1], such a pair can be understood so that $\#t$ represents the amount of information in favour of a statement and $\#f$ the amount of information against it. Hence, the truth value $(1,0)$ stands for "true", in the sense that there is information only in favour of the statement; $(0,1)$ stands for "false", information only against it; $(1,1)$ stands for both ("contradiction"); and $(0,0)$ stands for neither ("vagueness"). These values can be ordered by increasing truth $\leqslant_t$ and decreasing knowledge $\leqslant_k$ as shown in the diagrams below:



Lattice $\mathcal{FOUR}$ with $\leqslant_k$-order



Lattice $\mathcal{FOUR}$ with $\leqslant_t$-order

The framework underlying this paper generalizes $\mathcal{FOUR}$ by allowing weights to be parameterized over a class of lattices. Given a complete lattice $(A, \leqslant)$, one can construct a complete lattice A^2 as the product of A with its order dual A^∂. This construction, known as a *twist structure*, a term coined by Kracht [37], has been extensively studied in the literature [31, 35, 37]. In particular, Fitting [23] showed that it provides a general method for constructing bilattices, i.e., structures $(B, \leqslant_t, \leqslant_k, \neg)$ equipped with two orders $\leqslant_t$ and $\leqslant_k$, such that $(B, \leqslant_t)$ and $(B, \leqslant_k)$ are complete lattices, and $\neg$ is a unary involutive operator on B that is increasing w.r.t. $\leqslant_t$ and decreasing w.r.t. $\leqslant_k$ [1]. The notion of a logical bilattice and the corresponding (paraconsistent) logic system was introduced by Arieli and Avron [1] with applications to preference modelling [22].

Our line of research [15, 16, 18] focuses on *square bilattices* [22], where values are interpreted as pairs $(\#t, \#f) \in A^2$, where $\#t$ represents the amount of belief on an assertion and $\#f$ the degree of belief against it. For example, suppose that weights for both transitions come from a complete lattice over the interval $[0,1]$, see Figure 1. Such pairs of weights capture different behaviors:

- **Inconsistency**, when the positive and negative weights are contradictory, i.e., they sum to a value greater than 1, which corresponds to the upper triangle in Figure 1.
- **Vagueness**, when the sum is less than 1, corresponding to the lower triangle in Figure 1.

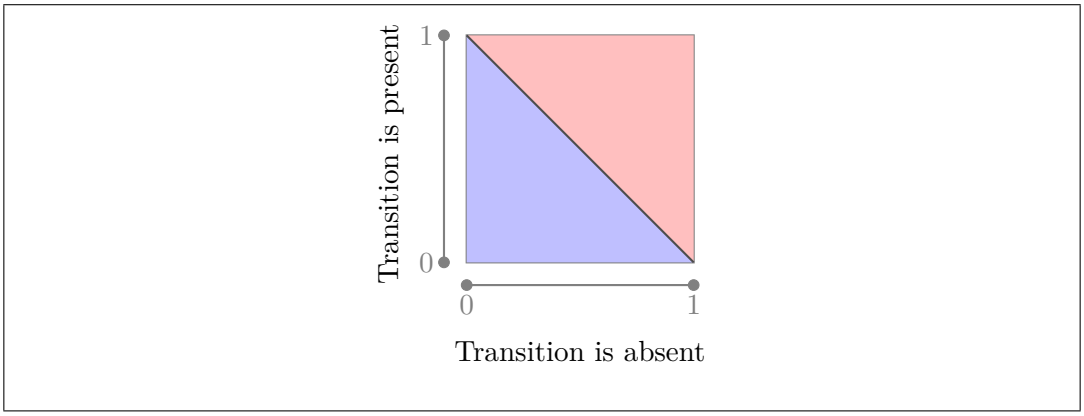


Figure 1: The vagueness-inconsistency square

- **Consistency**, when the sum is exactly 1, which means that the belief for and against an assertion are complementary, corresponding to the anti-diagonal line in Figure 1.

While fuzzy approaches are well-suited to represent vague information, our paraconsistent framework additionally accommodates inconsistencies. For instance, as discussed in research on bilattices applied to preference modeling [22], let $P(u, v)$ be a relation that expresses whether u is (known to be) at least as good as v. In a fuzzy setting, assigning $P(u, v) = 0$ may either mean that u is definitely not preferred to v, or that there is simply no information available to make a comparison. This ambiguity motivates the use of a pair of values rather than a single fuzzy value, in order to distinguish more precisely between different informational scenarios: *truth*, *falsity*, *ignorance*, and *contradiction*.

In many contexts, contradictions arise naturally from conflicting evidence, misinformation or evolving beliefs, and often go hand in hand with dynamic behavior. A legal trial is a classic example, frequently used in research in discursive logic, see [40, Section 4.2.]. In such settings, multiple witnesses may offer individually consistent testimonies that collectively conflict. Expert witnesses, due to their specialized knowledge, may refute earlier claims, thereby influencing which conclusions modeled as transitions (i.e., edges in the model) are considered reliable. Over time, trust in different sources may shift. For instance, if conflicting testimonies are given about a weapon and an expert rules out some possibilities, their input should affect the system by activating or deactivating certain transitions. Moreover, a witness credibility is not fixed: if a statement is discredited, future claims from that source may carry less weight. These evolving states highlight the natural connection between para-

consistency and dynamic behavior. A suitable modeling framework must therefore accommodate contradictions while adapting as new information becomes available.

Prior work [11] introduced paraconsistency into reactive graphs by employing Belnap's four-valued propositional variables. In addition, they proposed a paraconsistent logic and explored related topics, such as the measurement of inconsistencies, building on earlier contributions [12, 13]. In contrast, our approach incorporates paraconsistency directly into the edges of the graph, rather than into the propositional variables. This is achieved by allowing each edge to be weighted by a pair of values $(\mathit{tt}, \mathit{ff}) \in 2^2$. It is convenient to think of the pair $(\mathit{tt}, \mathit{ff})$ as a pair of extreme truth values, where tt represents the truth value of the statement "the transition will occur," and ff represents the truth value of the statement "the transition will be prevented from occurring." As a result, this framework can naturally represent vague, consistent, or inconsistent information, as described above. Consequently, all the relevant constructions in our framework are parametric in a class of lattices $(A, \leqslant)$. This accommodates different instantiations depending on the set A of possible values best suited to model the problem at hand.

This paper begins by introducing the necessary background definitions required for the sequel, including those related to Reactive Graphs, paraconsistent graphs as introduced in [15], and *twist-structures*, which are essential for reasoning with pairs of values. As introduced in [17], we then combined the latter graph notions to define the concept of a *Reactive Graph with Paraconsistent Transitions* (abbreviated as $\mathrm{RG_P}$). An $\mathrm{RG_P}$ is a graph in which ground transitions are weighted by pairs of truth values representing evidence for and against the occurrence of a transition, and where bivalent hyper-edges act as switches that activate or deactivate other edges. Additionaly in the original paper that this work extends [17], motivated by the work in [42], we presented a semantics for $\mathrm{RG_P}$ that supports the "unfolding" [30] of an $\mathrm{RG_P}$ into a paraconsistent graph, and provided a concrete application of the model. Building on this earlier conference paper [17], we extend this framework to introduce paraconsistent reactive graphs (abbreviated as PRG), in which both ground and hyper-edges are weighted by pairs of values. This generalization reflects the idea that the influence of hyper-edges may itself be uncertain or contradictory, rather than strictly bivalent.

In this paper, we begin by studying PRG under the assumption that hyper-edges are non-conflicting, i.e., no two hyper-edges are triggered by the same edge and act upon the same target. This restriction allows us to understand more precisely the impact that activating and deactivating hyper-edges have on the system's behavior. This extension requires a careful revision of the definitions originally presented in [17].

In a subsequent section, we address scenarios in which conflicting hyper-edges may arise. Since edges may encode information from diverse sources, it is natural to encounter situations where some sources support the activation of a transition while others support its deactivation. This raises important questions concerning conflict management within paraconsistent graphs. Moreover, throughout the paper, we explore results concerning consistency preservation under the proposed semantics.

In conclusion, we extend the work presented in the original conference paper [17] with the following main contributions:

1. We define paraconsistent reactive graphs, in which both ground and hyper-edges can represent vague, consistent or inconsistent information, extending the reactive graphs with paraconsistent transitions in [17], in which only the ground edges can represent non-bivalent information.

2. We develop a semantics for paraconsistent reactive graphs, first without conflicting hyper-edges, and then for the general case.

3. We prove monotonicity and consistency-preserving results for both semantics, with the latter following as a consequence of the novel Theorems 1 and 2.

4. We propose a number of conflict resolution policies, each supported by a clear motivation.

The remainder of the paper is organized as follows: Section 2 presents background definitions on reactive graphs, paraconsistent graphs and twist constructions which are essential for the sequel. Section 3 recalls the definition of Reactive Graphs with Paraconsistent transitions and their semantics, as introduced in the original conference paper [17]. Section 4 introduces paraconsistent reactive graphs and their semantics. The later section is divided in two: Section 4.1 outlines the formalism and results whenever paraconsistent reactive graphs have no conflicting hyper-edges and Section 4.2 addresses conflicts. Finally, Section 5 offers concluding remarks and identifies several lines for future research.

2 Preliminaries

We begin this section by recalling the definition of a *reactive graph*, as presented in [42]. We then present twist-structures as a construction parameterized by a class of lattices $(A, \leqslant)$ that enables the computation of pairs of values in A^2. Next, we recall the definition of a paraconsistent graph, introduced in [15]. Finally, in order to give

a concrete meaning to the vagueness-inconsistency square depicted in Figure 1 we endow the underlying lattice $(A, \leqslant)$ with a metric d, [14, 18].

The main goal of this section is to lay the groundwork for the subsequent sections, which aim to combine reactive graphs (see Definition 1) and paraconsistent graphs (see Definition 4) into a unique structure that captures the dynamic behavior of the former and the representation of inconsistencies in the latter.

2.1 Reactive Graphs

In this section, we recall the definition of a reactive graph as introduced in [42], along with its operational semantics.

Reactive graphs were proposed in [42] as an extension of reactive graphs originally introduced by Gabbay [27]. A reactive graph is a graph in which transitions between states, called ground edges ($\rightarrow$), occur via the execution of an action. In addition to ground edges, these graphs are also endowed with higher-order edges, known as hyper-edges, that represent the system's reactive behavior. Hyper-edges connect one edge to another and act as switches: whenever an edge is traversed, it may trigger a hyper-edge that either activates ($\twoheadrightarrow$) or deactivates ($\rightarrowtail$) another.

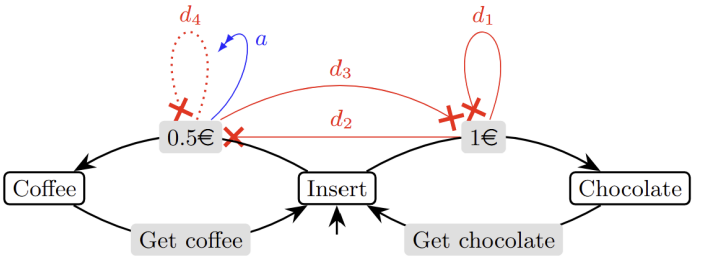
Definition 1. *Let Σ be a set of action symbols. A Σ-reactive graph is a tuple $M = (W, w_0, E, \mathcal{C}_0, \rightarrow, \twoheadrightarrow, \rightarrowtail, \dot{\cdot})$ where*

- *W is the set of states and $w_0 \in W$ is the initial state;*
- *E is the set of edge names and $\mathcal{C}_0 \subseteq E$ is the set of initially active edges;*
- *$\rightarrow \subseteq W \times \Sigma \times W$ is the set of ground edges; $\twoheadrightarrow \subseteq E \times E$ is the set of activating edges; $\rightarrowtail \subseteq E \times E$ is the set of deactivating edges;*
- *$\dot{\cdot} : E \longrightarrow (\rightarrow \dot{\cup} \twoheadrightarrow \dot{\cup} \rightarrowtail)$ is a bijective function that maps edges in E to their definition.*

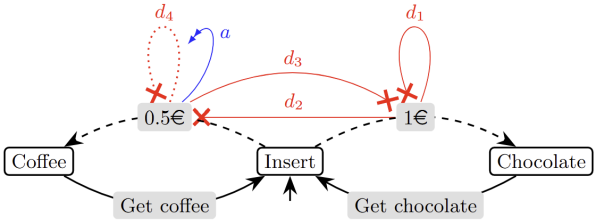
Graphically, edges that are not active are represented by a dashed arrow.

The following example illustrates how a vending machine can be modeled by a Σ-reactive graphs as defined above.

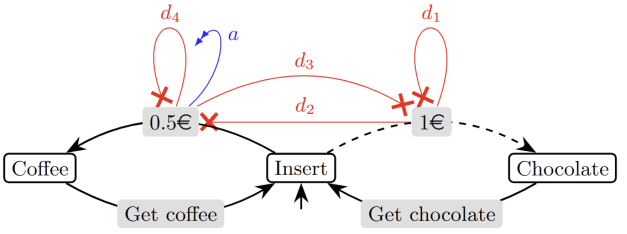
Example 1 ([42])**.** *The following reactive graph models the behavior of a buyer interacting with a vending machine that offers two products: coffee and chocolate. The buyer has only 1 euro to spend and can either purchase two coffees (at 0.5€ each) or one chocolate (at 1€).*



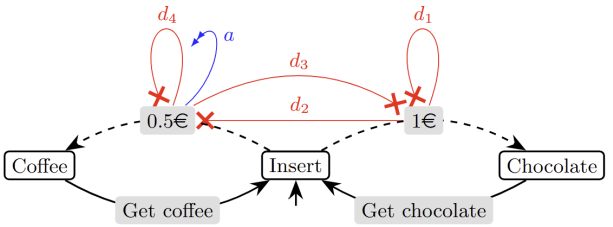
Intuitively, in the initial state, if the buyer performs the action 1€*, he immediately spends all the available funds and buys a chocolate. This triggers the action of deactivating edges* d_1 *and* d_2*, which disable any further action from the initial state. As a result, although the system returns to the initial state after the buyer gets a chocolate, no additional actions are available. Graphically, this is reflected by the fact that only dashed arrows, corresponding to inactive edges, originate from the initial state.*



Alternatively, if the buyer first performs the action 0.5€*, this triggers the deactivation edge* d_3*, which disables the edge labeled* 1€*, preventing the buyer from purchasing a chocolate since there are insufficient funds. This action also triggers the activation edge* a*, which activates edge* d_4*, allowing the buyer to perform only one more* 0.5€ *action, thus enabling the purchase of a second coffee. Graphically, after executing the action* 0.5€ *once, the system appears as follows:*



The buyer decision to purchase a second coffee, triggers edge d_4*, which disables any further coffee purchases, as the buyer has run out of funds. Graphically:*



Formally, we define this as a Σ*-reactive graph* $M = (W, w_0, E, \mathcal{C}_0, \rightarrow, \twoheadrightarrow, \multimap, \bar{\cdot})$*, where* $\Sigma = \{1€, 0.5€, \textit{Get coffee}, \textit{Get chocolate}\}$ *is the set of action symbols;* $W = \{\textit{Insert}, \textit{Coffee}, \textit{Chocolate}\}$ *is the set of states; and* $w_0 = \textit{Insert}$ *is the initial state. The set of edge names is* $E = \{1€, 0.5€, \textit{Get coffee}, \textit{Get chocolate}, d_1, d_2, d_3, d_4, a\}$*. We slightly abuse notation by naming the ground edges after their corresponding action symbols to improve readability.* $\mathcal{C}_0 = E \backslash \{d_4\}$ *denotes the set of initially active edges. The set of ground edges* $\rightarrow$ *are those depicted in black in the graphs; the activating edges* $\twoheadrightarrow$ *are depicted in blue; and the deactivating edges* $\multimap$ *are shown in red. Finally, the function* $\bar{\cdot}$ *defines the correspondence between edge names and their internal behavior. For example, the ground edge named* $0.5€$ *is mapped to* $\langle \textit{Insert}, 0.5€, \textit{Coffee} \rangle$*, indicating that this ground edge goes from state Insert to Coffee by the action* $0.5€$*. While the deactivation edge* d_2 *is mapped to its source and target, that is,* $\overline{d_2} = \langle 1€, 0.5€ \rangle$*, and similarly, the activation edge* a *is mapped as* $\overline{a} = \langle 0.5€, d_4 \rangle$*.*

Next we recall a few concepts originally introduced in [42], which play a crucial role in defining the operational semantics of reactive graphs.

Given a reactive graph M, an edge $e_s \in E$, and the current set of active edges $\mathcal{C} \subseteq E$. Then, $\mathsf{from}(e_s)$ denotes the set of hyper-edges that directly originate from e_s. Additionally, $\mathsf{from}^*(e_s, \mathcal{C})$ denotes the set that recursively transverses from to collect all active hyper-edges,i.e., edges in $\mathcal{C}$, triggered by e_s.

$$\begin{aligned} \mathsf{from}(e_s) &= \{e \mid \exists e_t \cdot \overline{e} = (e_s, e_t)\} \\ \mathsf{from}^*(e_s, \mathcal{C}) &= \bigcup_{r \in (\mathsf{from}(e_s) \cap \mathcal{C})} \mathsf{from}^*(r, \mathcal{C} \backslash \{e_s\}) \cup \{r\} \end{aligned} \tag{1}$$

where $\mathsf{from}^\star(r, \emptyset) = \emptyset$ for any $r \in E$.

Additionally, the set of edges activated by some edge e (respectively deactivated by e), written $\mathsf{on}(e, \mathcal{C})$ (respectively $\mathsf{off}(e, \mathcal{C})$) is defined as follows [42].

$$\begin{aligned} \mathsf{on}(e, \mathcal{C}) &= \{e_t \mid e_{trg} \in \mathsf{from}^*(e, \mathcal{C}) \wedge \exists e_s \cdot \overline{e_{trg}} = (e_s, e_t) \in \twoheadrightarrow\} \\ \mathsf{off}(e, \mathcal{C}) &= \{e_t \mid e_{trg} \in \mathsf{from}^*(e, \mathcal{C}) \wedge \exists e_s \cdot \overline{e_{trg}} = (e_s, e_t) \in \multimap\} \end{aligned} \tag{2}$$

Given a reactive graph, it is possible to unfold its reactive behavior into an ordinary graph—that is, a graph with no hyper-edges. In fact, Gabbay's work introduced not only an unfolding method that transforms a reactive graph into a standard one, but also a folding method that reverses this transformation. Together, these techniques show the expressive power of reactive graphs: folding can exponentially reduce the number of states of the graph, while unfolding enables the application of standard techniques, such as those used in the axiomatization of logics over reactive frames [29].

In this paper, we focus exclusively on the unfolding process that transforms a reactive graph into a standard graph. Before introducing this transformation through the operational semantics presented in Definition 2, we first formally define what it means for a reactive graph to evolve from one configuration to another.

For a Σ-reactive graph $M = (W, w_0, E, \mathcal{C}_0, \rightarrow, \twoheadrightarrow, \nrightarrow, \bar{\cdot})$, a *configuration* is a tuple $\langle w, \mathcal{C} \rangle$, where $w \in W$ is a state and $\mathcal{C} \subseteq E$ is a set of active edges.

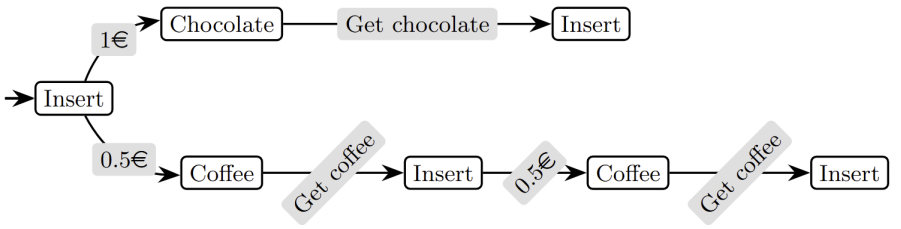
Given the inherently dynamic behavior of reactive graphs, in which traversing some ground edges may activate or deactivate other's, the system can evolve from an initial configuration $\langle w_0, \mathcal{C}_0 \rangle$ to a new one $\langle w, \mathcal{C} \rangle$. We refer to each such transition, from one configuration to another, as an evolution. Intuitively, an evolution records how the system changes after a single traversal of a ground edge. Formally, given two configurations $\langle w, \mathcal{C} \rangle$ and $\langle w', \mathcal{C}' \rangle$, we say that $\langle w, \mathcal{C} \rangle$ evolves to $\langle w', \mathcal{C}' \rangle$ if there exists an active ground edge connecting w to w', and the set of active edges is updated from $\mathcal{C}$ to $\mathcal{C}'$ after traversing that edge. This notion provides the foundation for the unfolding process, which transforms a reactive graph into a standard one by systematically considering all possible evolutions starting from the initial configuration.

Definition 2 ([42])**.** *The semantics of a graph M results from the evolution of the initial configuration $\langle w_0, \mathcal{C}_0 \rangle$, given by the rule below.*

$$\frac{\exists e \in \mathcal{C} \ \cdot \ \bar{e} = (w, a, w') \ \wedge \ \mathcal{C}' = \big(\mathcal{C} \cup \mathsf{on}(e, \mathcal{C})\big) \backslash \mathsf{off}(e, \mathcal{C})}{\langle w, \mathcal{C} \rangle \xrightarrow{a}_M \langle w', \mathcal{C}' \rangle}$$

where $\langle w, \mathcal{C} \rangle$ and $\langle w', \mathcal{C}' \rangle$ are states of M and there is a transition in M from $\langle w, \mathcal{C} \rangle$ to $\langle w', \mathcal{C}' \rangle$ triggered by action a.

Example 2 ([42])**.** *Using the semantics defined in Definition 2 to the reactive graph in Example 1 we obtain the following graph.*



Note that each state in the above standard graph is of the form $\langle w, \mathcal{C} \rangle$, where $\mathcal{C}$ is the set of active edges updated by the traversal of an edge. However, to improve readability, configurations are omitted from the state labels.

2.2 Paraconsistent Graphs

Before introducing the notion of a paraconsistent graph parameterized by a class of lattices, let us review the underlying semantic structures.

As in recent works on many-valued frameworks [6], our approach is parameterized over a class of complete lattices. Each lattice $\mathbf{A} = (A, \leqslant)$ acts over a non-empty set A of possible values, equipped with binary operations $\sqcap$ (meet) and $\sqcup$ (join) satisfying the usual distributive laws.

Example 3. *Well-known examples of lattices used throughout this paper include*[1]*:*

- *The two element Boolean algebra $\mathbf{2} = (\{0, 1\}, \leqslant_{\mathbf{2}})$ where $0 \leqslant_{\mathbf{2}} 1$, and the meet and join operations are standard Boolean conjunction and disjunction.*
- *The Łukasiewicz three-valued algebra $\mathbf{3} = (\{1, u, 0\}, \leqslant_{\mathbf{3}})$ where $0 \leqslant_{\mathbf{3}} u \leqslant_{\mathbf{3}} 1$. The intermediate value u stands for "unknown" and is also denoted in the literature by $1/2$ or $\#$, often interpreted as "possibility" or "indeterminacy" [40]. The meet and join operations are defined as follows:*

$\sqcap_{\mathbf{3}}$	0	u	1
0	0	0	0
u	0	u	u
1	0	u	1

$\sqcup_{\mathbf{3}}$	0	u	1
0	0	u	1
u	u	u	1
1	1	1	1

- *The Gödel algebra $\mathbf{G} = ([0, 1], \leqslant_{\mathbf{G}})$ where $\sqcap$ and $\sqcup$ correspond to the minimum and maximum functions on $[0, 1]$, respectively. For any $x, y \in [0, 1]$, $x \leqslant_{\mathbf{G}} y$ if and only if $x = \min(x, y)$.*

[1] Indices are omitted when clear from context.

As discussed above, this framework was extended in in [15, 18] by working with pairs of values $(\mathit{tt}, \mathit{ff}) \in A^2$, parameterized over a lattice $\mathbf{A} = (A, \leqslant)$, which enables the representation of vague, consistent, and inconsistent information. Here, tt denotes the degree of belief in favor of a statement, while ff denotes the degree of belief against it. These pairs form a twist-structure [23], also known as a square bilattice [2], which provides a systematic way to define operations on A^2.

Definition 3. *Let $\mathbf{A} = (A, \leqslant)$ be a complete distributive lattice over a non-empty set of values A. The structure $\mathcal{A} = (A \times A, \leqslant_t, \leqslant_k, \neg)$ is $\mathbf{A}$-twist structure where $\neg(\mathit{tt}, \mathit{ff}) = (\mathit{ff}, \mathit{tt})$ and*

- $(\mathit{tt}_1, \mathit{ff}_1) \leqslant_t (\mathit{tt}_2, \mathit{ff}_2)$ *if and only if* $\mathit{tt}_1 \leqslant \mathit{tt}_2$ *and* $\mathit{ff}_1 \geqslant \mathit{ff}_2$
- $(\mathit{tt}_1, \mathit{ff}_1) \leqslant_k (\mathit{tt}_2, \mathit{ff}_2)$ *if and only if* $\mathit{tt}_1 \leqslant \mathit{tt}_2$ *and* $\mathit{ff}_1 \leqslant \mathit{ff}_2$

Denoting the join and meet operations of $\mathbf{A}$ by $\sqcap$ and $\sqcup$ respectively, the following operators are defined for any pair $(\mathit{tt}_1, \mathit{ff}_1), (\mathit{tt}_2, \mathit{ff}_2) \in A^2$:

$$
\begin{aligned}
(\mathit{tt}_1, \mathit{ff}_1) \sqcap (\mathit{tt}_2, \mathit{ff}_2) &= (\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcup \mathit{ff}_2) \\
(\mathit{tt}_1, \mathit{ff}_1) \sqcup (\mathit{tt}_2, \mathit{ff}_2) &= (\mathit{tt}_1 \sqcup \mathit{tt}_2, \mathit{ff}_1 \sqcap \mathit{ff}_2) \\
(\mathit{tt}_1, \mathit{ff}_1) \otimes (\mathit{tt}_2, \mathit{ff}_2) &= (\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcap \mathit{ff}_2) \\
(\mathit{tt}_1, \mathit{ff}_1) \oplus (\mathit{tt}_2, \mathit{ff}_2) &= (\mathit{tt}_1 \sqcup \mathit{tt}_2, \mathit{ff}_1 \sqcup \mathit{ff}_2)
\end{aligned}
$$

The order relation $\leqslant$ in $\mathbf{A}$ induces two orders on A^2: the truth order $\leqslant_t$ and the knowledge order $\leqslant_k$. Operations $\sqcap$ and $\sqcup$ correspond to the meet and join under $\leqslant_t$, intuitively representing conjunction ("and") and disjunction ("or"), respectively. Operations $\otimes$ and $\oplus$ correspond to meet and join under $\leqslant_k$ and have distinct epistemic interpretations [1]. Specifically, $\otimes$ represents consensus, so that $p \otimes q$ yields the highest amount of information on which both p and q agree with respect to the knowledge order $\leqslant_k$, while $\oplus$ can be viewed as guillibity or "credulousness", accepting the combined knowledge provided by p and q. For instance, $(1, 0) \otimes (0, 1) = (0, 0)$, since one value represents complete belief in a statement, $(1, 0)$, and the other complete disbelief, $(0, 1)$; so the most they can agree on is nothing, $(0, 0)$. In contrast, $(1, 0) \oplus (0, 1) = (1, 1)$, as the combination of both values represents simultaneous belief and disbelief, resulting in a contradiction. Finally, the unary operator $\neg$ corresponds to an involutive negation [1]. Intuitively, if the evidence for and against a statement is tt and ff, respectively, then the evidence for and against its negation is ff and tt, respectively. As expected, this negation reverses truth; that is, if $(\mathit{tt}, \mathit{ff}) \leqslant_t (\mathit{tt}', \mathit{ff}')$, then $\neg(\mathit{tt}, \mathit{ff}) \geqslant_t \neg(\mathit{tt}', \mathit{ff}')$. Moreover, it preserves knowledge: if $(\mathit{tt}, \mathit{ff}) \leqslant_k (\mathit{tt}', \mathit{ff}')$, then $\neg(\mathit{tt}, \mathit{ff}) \leqslant_k \neg(\mathit{tt}', \mathit{ff}')$, so that one

knows neither more nor less about a statement than about its negation [24]. These operators play a central role in the Section 4, to combine information from multiple edges in paraconsistent graphs.

Since $\mathbf{A} = (A, \leqslant)$ is distributive, so is the corresponding twist-structure $\mathcal{A} = (A \times A, \leqslant_t, \leqslant_k, \neg)$ [22, Proposition 2]. That is, all twelve distributive laws involving $\sqcap$, $\sqcup$, $\otimes$, and $\oplus$ hold. For example, for any values $(\mathit{tt}_1, \mathit{ff}_1)$, $(\mathit{tt}_2, \mathit{ff}_2)$ and $(\mathit{tt}_3, \mathit{ff}_3)$ in A^2, the following distributive law holds:

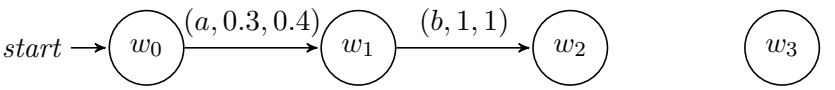
$$((\mathit{tt}_1, \mathit{ff}_1) \sqcap (\mathit{tt}_2, \mathit{ff}_2)) \otimes (\mathit{tt}_3, \mathit{ff}_3) = ((\mathit{tt}_1, \mathit{ff}_1) \otimes (\mathit{tt}_3, \mathit{ff}_3)) \sqcap ((\mathit{tt}_2, \mathit{ff}_2) \otimes (\mathit{tt}_3, \mathit{ff}_3))$$

As previously stated, this framework underpins our earlier work [15, 18], where we introduced the notion of a paraconsistent graph, i.e., a graph in which transitions are labeled by pairs of values in A^2. Many-valued reasoning is often parametric on lattices (see, e.g., [6, 33, 38]) or bilattices (see, e.g., [1, 24]), which makes it convenient to refer to their elements as truth values. Accordingly, the pairs $(\mathit{tt}, \mathit{ff}) \in A^2$ can be viewed as representing a space of truth values, intuitively capturing the extent to which the statement "the transition is possible" holds or not, respectively. Let us recall the formal definition of a paraconsistent graph.

Definition 4. *Let $(A, \leqslant)$ be a complete distributive lattice and Σ a set of action symbols. An (A, Σ)-paraconsistent graph is a tuple $M = (W, w_0, R)$ where*

- *W is a non-empty set of states and $w_0 \in W$ is the initial state;*
- *$R \subseteq W \times \Sigma \times A^2 \times W$ is a paraconsistent relation, i.e., if $(w, a, \mathit{tt}, \mathit{ff}, w') \in R$, then tt measures the evidence of the transition from w to w' occurring via action a and ff measures the evidence of the transition being prevented from occurring.*

Example 4. *The structure $(\{w_0, w_1, w_3\}, w_0, R)$ depicted below is a $([0,1], \{a, b\})$-paraconsistent graph, whose transitions are labeled by pairs of truth values in $[0,1]^2$, capturing the extent to which each transition may occur or be prevented from occurring.*



Transition $(w_0, a, 0.3, 0.4, w_1)$ represents vague information, since the sum of weights is less than one, $0.3 + 0.4 < 1$. In contrast, $(w_1, b, 1, 1, w_2)$ represents inconsistent information, as the sum of the weights exceeds 1. Transitions that are known to be stricly false, such as $(w_2, a, 0, 1, w_3)$, are omitted from the diagram for clarity.

Finally, as pointed out in the Introduction, in order to precisely describe the two triangular regions shown in Figure 1 and the diagonal separating them, we endow the underlying lattice $\mathbf{A}$ with a metric $d : A \to \mathbb{R}$.

Example 5. *For each of the lattices in Ex. 3, we provide an suitable choice of d.*

For the Boolean algebra $\mathbf{2}$*, the three valued algebra* $\mathbf{3}$ *and the Gödel algebra* $\mathbf{G}$ *consider the following metrics, respectively.*

$$d_{\mathbf{2}}(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{otherwise} \end{cases} \qquad \begin{array}{c|ccc} d_{\mathbf{3}} & 0 & u & 1 \\ \hline 0 & 0 & 1 & 2 \\ u & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{array} \qquad d_{\mathbf{G}}(x,y) = \sqrt{(x-y)^2}$$

Each metric d can be lifted to a metric $D : A^2 \to \mathbb{R}$ defined over pairs of values, as expected:

$$D((\mathit{tt}_1, \mathit{ff}_1), (\mathit{tt}_2, \mathit{ff}_2)) = \sqrt{d(\mathit{tt}_1, \mathit{tt}_2)^2 + d(\mathit{ff}_1, \mathit{ff}_2)^2} \tag{3}$$

This metric is particularly relevant as it provides a concrete interpretation for the vagueness-inconsistency square depicted in Figure 1. Consequently, it enables the definition of the consistent and paraconsistent sets, introduced in [15], denoted by Δ_C and Δ_P, respectively.

$$\begin{aligned} \Delta_C &= \{(\mathit{tt}, \mathit{ff}) \mid D((\mathit{tt}, \mathit{ff}), (0,0)) \leqslant D((\mathit{tt}, \mathit{ff}), (1,1))\} \\ \Delta_P &= \{(\mathit{tt}, \mathit{ff}) \mid D((\mathit{tt}, \mathit{ff}), (1,1)) < D((\mathit{tt}, \mathit{ff}), (0,0))\} \end{aligned} \tag{4}$$

The set Δ_C encompasses all pairs within the lower triangle and the black anti-diagonal line in Figure 1, while Δ_P includes pairs located within the upper triangle. Therefore, by comparing the distance of any pair of weights to the elements $(0,0)$ and $(1,1)$, it is possible to determine whether it represents consistent or inconsistent information. Moreover, the set $\Delta = \{(\mathit{tt}, \mathit{ff}) \mid D((\mathit{tt}, \mathit{ff}), (0,0)) = D((\mathit{tt}, \mathit{ff}), (1,1))\}$ includes all pairs equidistant from $(0,0)$ and $(1,1)$, representing *strictly* consistent information.

3 Reactive Graphs with Paraconsistent Transitions

This section revisits the definition of reactive graphs with paraconsistent transitions, introduced in the original conference paper that this work extends [17]. In this framework dynamic behavior is modeled through hyper-edges that serve as switches, either activating ($\twoheadrightarrow$) or deactivating ($\nrightarrow$) other edges. Although such hyper-edges are not themselves traversable, they encapsulate the system's reactivity by indicating

how edges change in response to the traversal of a ground edge ($\rightarrow$). Paraconsistency is incorporated solely in ground edges, which are weighted by pairs of values in A^2 over a lattice $\mathbf{A} = (A, \leqslant)$.

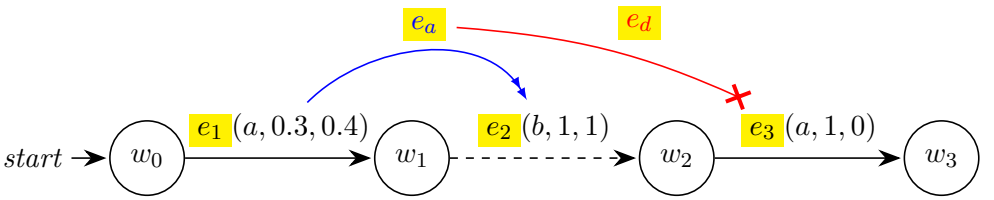
In this first step, our aim is to combine reactive graphs (see Definition 1) and paraconsistent graphs (see Definition 4) into a novel structure that captures the dynamic behavior of the former and the representation of inconsistencies in the latter. In the next section, we will extend this definition to paraconsistent reactive graphs, where all edges ($\rightarrow$, $\twoheadrightarrow$, $\rightarrowtail$) are weighted by a pair of values in A^2. This generalization is motivated by the observation that switches themselves may carry vague or inconsistent information, arising from incomplete knowledge or internal contradictions. This, in turn, influences the system's behavior.

The definitions presented in this section are adapted from recent work on reactive graphs [42]. We define reactive graphs with paraconsistent transitions as an extension of paraconsistent graphs with hyper-edges. Then, their operational semantics is obtained by "unfolding" [29] them back into standard paraconsistent graphs. This process effectively eliminates hyper-edges while preserving the overall behavior. We conclude the section with a worked example.

Definition 5. *Let $(A, \leqslant)$ be a complete distributive lattice and Σ a set of action symbols. An (A, Σ)-reactive graph with paraconsistent transitions, abbreviated to* $\mathrm{RG_P}$*, is a tuple $M = (W, w_0, \mathcal{C}_0, E, \rightarrow, \twoheadrightarrow, \rightarrowtail, \bar{\cdot})$ where:*

- *W is a set of states and $w_0 \in W$ is the initial state;*
- *E is the set of all edge names and $\mathcal{C}_0 \subseteq E$ is the set of initially active edges;*
- *$\rightarrow \subseteq W \times \Sigma \times A^2 \times W$ is the set of ground edges, $\twoheadrightarrow \subseteq E \times E$ is the set of activating edges and $\rightarrowtail \subseteq E \times E$ is the set of deactivating edges;*
- *$\bar{\cdot} : E \longrightarrow (\rightarrow \dot{\cup} \twoheadrightarrow \dot{\cup} \rightarrowtail)$ is a function that maps an edge in E to its definition*

Example 6. *The system depicted below is a $([0,1], \{a, b\})$-$\mathrm{RG_P}$. The set of all edges is $E = \{e_1, e_2, e_3, e_a, e_d\}$, each marked in yellow in the diagram below. The initial set of active edges is given by $\mathcal{C}_0 = E \backslash \{e_2\}$. As before, the inactive edge e_2 is depicted as a dashed arrow. The function $\bar{\cdot}$ maps each edge to its internal details. For instance, for the ground edge e_3 and the hyper-edge e_d, we have $\overline{e_3} = (w_2, a, 1, 0, w_3)$ and $\overline{e_d} = (e_a, e_3)$.*



Note that hyper-edges act purely as switches and have no direct impact on the weights of ground transitions.

For an (A, Σ)-RG$_\mathrm{P}$, let $\mathcal{C} \subseteq E$ be the current set of active edges, and $e_s \in E$ be some edge. As before, $\mathsf{from}(e_s)$ and $\mathsf{from}^\star(e_s, \mathcal{C})$ denote the sets of hyper-edges triggered by e_s, either directly or recursively, as defined in (1). Additionally, the set of edges activated (resp. deactivated) by e_s, denoted $\mathsf{on}(e_s, \mathcal{C})$ (resp. $\mathsf{off}(e_s, \mathcal{C})$), is defined as in (2). The following example illustrates these definitions in the context of an RG$_\mathrm{P}$.

Example 7. *Recall Example 6. Edge e_1 represents a transition from state w_0 to w_1 via action a, with degrees of belief 0.3 for occurrence and 0.4 for non-occurrence. Hence, we have $\overline{e_1} = (w_0, a, 0.3, 0.4, w_1)$. This edge triggers the hyper-edges e_a and e_d, which in turn activate e_2 and deactivate e_3, respectively. We can now compute:*

- $\mathsf{from}(e_1) = \{e_a\}$ *returns the hyper-edges directly triggered by e_1;*
- $\mathsf{from}^*(e_1, \mathcal{C}_0) = \{e_a, e_d\}$ *returns all the hyper edges triggered by e_1 under the initial set of active edges $\mathcal{C}_0$;*
- $\mathsf{on}(e_1, \mathcal{C}_0) = \{e_2\}$ *and* $\mathsf{off}(e_1, \mathcal{C}_0) = \{e_3\}$ *return the edges targeted by activating and deactivating hyper-edges, respectively, triggered by e_1.*

The following definition introduces the operational semantics for the sort of graphs considered in this section. Given an RG$_\mathrm{P}$, this semantics unfolds its dynamic behavior into a paraconsistent graph without hyper-edges [29].

Definition 6. *The semantics of an (A, Σ)-paraconsistent graph P, as defined in Definition 4, results from the evolution of the initial configuration $\langle w_0, \mathcal{C}_0 \rangle$, given by the rule below.*

$$\frac{\exists e \in \mathcal{C} \;\cdot\; \overline{e} = (w, a, \mathit{tt}, \mathit{ff}, w') \;\wedge\; \mathcal{C}' = \big(\mathcal{C} \cup \mathsf{on}(e, \mathcal{C})\big) \backslash \mathsf{off}(e, \mathcal{C})}{\langle w, \mathcal{C} \rangle \xrightarrow{(a, \mathit{tt}, \mathit{ff})}_P \langle w', \mathcal{C}' \rangle}$$

where $\langle w, \mathcal{C} \rangle$ and $\langle w', \mathcal{C}' \rangle$ are states of P connected by a paraconsistent transition via the action symbol a and weighted by the pair $(\mathit{tt}, \mathit{ff})$.

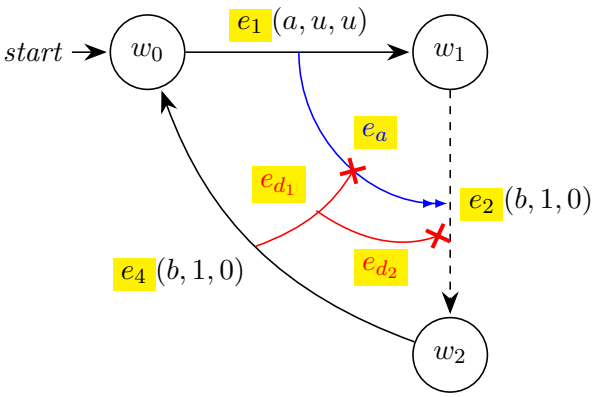
Applied to Example 6 this yields the standard paraconsistent graph in Example 4.

As in [42], this semantics differs from Gabbay's original one by collecting all activated edges before applying their (de)activation effects, instead of triggering effects immediately during traversal.

Additionally, this semantics introduces a bias: when an edge is both activated and deactivated within the same step, deactivation takes precedence. While this choice simplifies the semantics above, it may not always reflect the intended behavior. We will return to this issue in Section 4.2, when considering paraconsistent reactive graphs with conflicting hyper-edges and propose some policies for managing such conflicts.

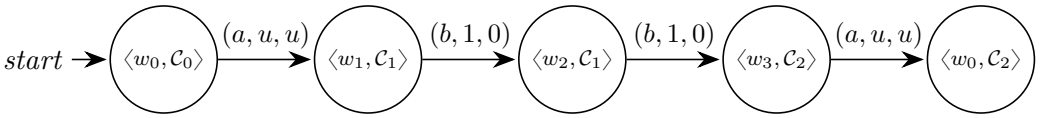
The following example illustrates the unfolding of a reactive graph with paraconsistent transitions.

Example 8. *Consider the following* $(\{a, b\}, \{0, u, 1\} - \mathrm{RG_P}$ *where traversing edge* e_1 *triggers the hyper-edge* e_a *that activates* e_2*, and traversing edge* e_4 *triggers the hyper-edges* e_{d_1} *and* e_{d_2} *that deactivate* e_a *and* e_2*, respectively.*



Essentially, we can follow the path $w_0 \xrightarrow{e_1} w_1 \xrightarrow{e_2} w_2 \xrightarrow{e_4} w_0 \xrightarrow{e_1} w_1$*. By the time we return to* w_1*, transition* e_2 *is no longer active, and the hyper-edge that could activate it,* e_a*, has also been deactivated. Hence, there is no other active edge to traverse. More precisely, if we follow the path as described, the initial set of active edges* $\mathcal{C}_0 = \{e_1, e_4, e_a, e_{d_1}, e_{d_2}\}$ *is updated to* $\mathcal{C}_1 = \mathcal{C}_0 \cup \{e_2\}$ *when traversing edge* e_1*, since this edge triggers the activation of* e_2*. Subsequently, traversing edge* e_2 *has no impact on the set of active edges, so the set remains equal to* $\mathcal{C}_1$*. Then, traversing edge* e_4 *updates the active edges from* $\mathcal{C}_1$ *to* $\mathcal{C}_2 = \mathcal{C}_1 \backslash \{e_a, e_2\}$ *by deactivating* e_a *and* e_2*. Finally, traversing* e_1 *again does not trigger any further activation or deactivation.*

This corresponds exactly to the unfolding of the $\mathrm{RG_P}$ *as defined in Definition 15. Therefore, this* $\mathrm{RG_P}$ *can be unfolded into the following paraconsistent graph.*



Finally, we present a concrete application of reactive graphs with paraconsistent transitions.

Example 9 (Liar's Coin)**.** *Two players are engaged in a game called* Liar's Coin*, which is adapted from work on epistemic logic [39]. Player 1 bets on heads, while Player 2 bets on tails. Player 1 tosses a coin and announces either heads or tails, but keeps the actual outcome hidden from Player 2. If Player 1 announces tails* $(\dagger t)$*, Player 2 wins, and the game ends. If Player 1 announces heads* $(\dagger h)$*, Player 2 has two options: pass* (p)*– in which case Player 1 wins– or challenge* (c)*, forcing Player 1 to show the coin* (h *or* t)*. The player with the correct bet then wins. From Player 2's perspective, the game can be represented by the graph*[2] *in Figure 2.*

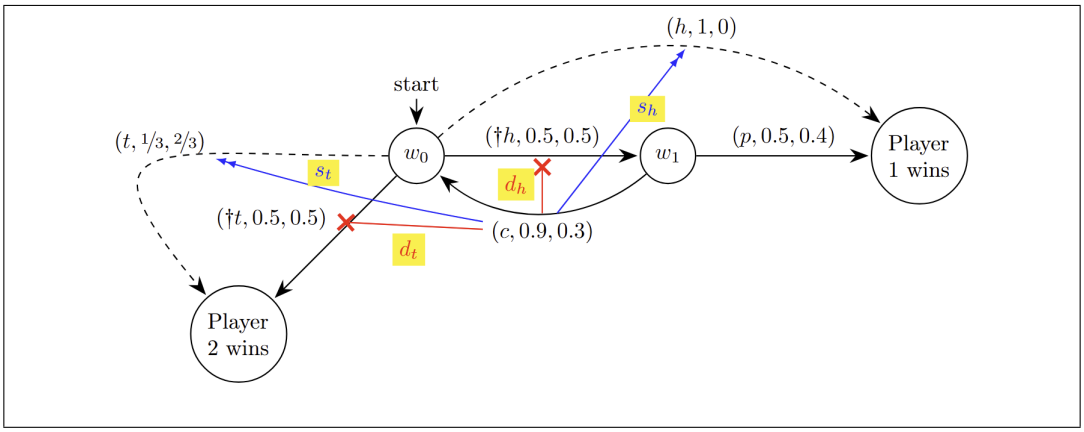


Figure 2

Initially, Player 2's belief about the coin outcome (heads or tails) reflects that of a fair coin toss. Therefore, the transitions representing Player 1 announcing heads or tails, $\dagger h$ *and* $\dagger t$*, are given by the consistent truth value* $(0.5, 0.5)$*, indicating equal uncertainty for and against the outcome.*

When Player 1 announces heads, Player 2 must choose to either challenge or pass. This decision may depend on various vague cues– Player 1's demeanor, prior experience, or intuition. If Player 2 chooses to challenge, the announcement edges,

[2] To improve readability, we omit the names of the ground edges, which are highlighted in yellow. For the original graph, see the conference paper that this work extends [17].

given by the actions †h and †t, are deactivated, and the edges given by the actions h and t are activated. These edges are assigned weights that reflect Player 1's likelihood of honesty. In this example, Player 1 is assumed never to lie about a heads result, so the ground-edge with action h has a value of $(1, 0)$*. For the ground edge with action t, however, the evidence is more uncertain: suppose from previous games Player 2 believes Player 1 lies about tails with probability* $1/3$*. Then, revealing tails is weighted with* $(1/3, 2/3)$*.*

Now suppose Player 1 announces "heads". Player 2 must evaluate the evidence for challenging or passing. This evidence may include:

1. *In the last two rounds, Player 1 also announced "heads", which seems unlikely.*
2. *Player 2 knows from prior rounds that Player 1 is generally honest.*
3. *Player 1 appeared nervous during the announcement.*
4. *Player 2 is a risk-taker and inclined to challenge.*

Based on this, Player 2 might subjectively assign non-complementary degrees of belief to each action. These values need not be complementary, as Player 2's reasoning integrates both statistical and intuitive arguments. Using the information from points (1)–(4)*, Player 2 ranks the evidence for and against challenging or passing as follows:*

- *Evidence to challenge is the strongest, say* 0.9*, supported by* (1)*,* (3) *and* (4)*.*
- *Evidence to pass follows, say* 0.5*, moderately supported by* (2) *but tempered by skepticism from* (1)*.*
- *Evidence against passing comes next, say* 0.4*, based partly on* (2) *and* (4)*. While both are vague,* (4) *plays a stronger role in Player 2's behavior.*
- *Evidence against challenging is the weakest, say* 0.3*, primarily based on* (2)*.*

Thus, the edges corresponding to the challenging (c) and passing (p) actions are assigned the values $(0.9, 0.3)$ *and* $(0.5, 0.4)$*, respectively.*

4 Paraconsistent Reactive Graphs

We now extend the definition of reactive graphs with paraconsistent transitions (Definition 5) by allowing hyper-edges ($\twoheadrightarrow$ and $\rightarrowtail$) to be also weighted by pairs of

truth values in A^2. As a result, all transitions in the graph, including both ground and hyper-edges, may now represent vague, consistent or inconsistent information.

This extension is motivated by scenarios in which information is provided by multiple, possibly unreliable or conflicting, sources. For example, recall the "Liars Coin" example (Example 9). Suppose there are now three players: Player 1 flips the coin and hides the result, while Player 3– whose intentions are unclear– observes the outcome and announces either "heads" or "tails," potentially siding with Player 1 or Player 2 depending on their own interests. In such a case, Player 2 receives contradictory information if Player 1 announces "heads" and Player 3 announces "tails," or vice versa, which can be represented by the pair $(1, 1)$. If Player 2 believes that Player 3's statement undermines Player 1's, they may be incentivized to distrust Player 1 and challenging him– i.e., activating the "challenge" edge. Such dynamics illustrates the need for non-classical hyper-edges that capture uncertainty or inconsistency in the information provided by agents.

This section is organized in two parts:

- Subsection 4.1, introduces paraconsistent reactive graphs (PRG) under the restriction that no conflicting hyper-edges are allowed. Specifically, no two hyper-edges may be triggered by the same source edge and target the same edge. This notion, mathematically formalized, contributes to the understanding of the isolated influence of paraconsistent hyper-edges in a controlled setting.
- Subsection 4.2,addresses the more general and challenging case in which conflicting hyper-edges are permitted. Then, we explore policies for resolving contradictions arising from different sources of information.

Let us now define paraconsistent reactive graphs.

Definition 7. *Let $(A, \leqslant)$ be a complete distributive lattice and Σ a set of action symbols. An (A, Σ)-paraconsistent reactive graph, abbreviated to* PRG*, is a tuple $M = (W, w_0, \rightarrow, \twoheadrightarrow, \rightarrowtail, \mathcal{C}_0, \bar{\cdot}, \bar{\bar{\cdot}})$ where:*

- *W is a set of states and $w_0 \in W$ is the initial state;*
- *$\rightarrow$, $\twoheadrightarrow$ and $\rightarrowtail$ are the sets of ground, activating, and deactivating edge names, respectively. We define $E = (\rightarrow \cup \twoheadrightarrow \cup \rightarrowtail)$ as the set of all edges, and $H = (\twoheadrightarrow \cup \rightarrowtail)$ as the set of all hyper-edges.*
- *$\mathcal{C}_0 : E \longrightarrow A^2$ is the initial activation function, assigning to each edge $e \in E$ a pair $(tt, ff) \in A^2$, where tt and ff represent the degrees of belief supporting and opposing the traversal of e, respectively.*

- $\dot{-} : \rightarrow \longrightarrow (W \times \Sigma \times W)$ *and* $\dot{\dot{-}} : H \longrightarrow E^2$ *are functions such that* $\dot{-}$ *maps each ground edge in* $\rightarrow$ *to its definition, while* $\dot{\dot{-}}$ *maps each hyper-edge to the pair of edges it connects: the trigger and the target.*

The notation adopted here is slightly different from that used in the previous section for reactive graphs with paraconsistent transitions (see Definition 5). This adjustment is necessary to accommodate the fact that all transitions, both ground edges and hyper-edges, are now weighted by pairs in A^2.

The main change is that the set of initial active edges $\mathcal{C}_0 \subseteq E$ is now replaced by an initial activation function $\mathcal{C}_0 : E \longrightarrow A^2$, which assigns to each edge a pair of values representing the positive and negative evidence for its execution. For example, if $\mathcal{C}_0(e_1) = (0, 1)$, this corresponds to the classical case in which edge e_1 is inactive. Visually, we will no longer use dashed arrows in diagrams to represent inactive edges, as the active/inactive edges are now determined by the values assigned by $\mathcal{C}_0$.

As it will become clear later, it is important to distinguish between ground, activating, and deactivating edges, especially since their values may evolve during execution as the behavior of one edge impacts the weight of others. For this reason, we explicitly define the sets $\rightarrow$, $\twoheadrightarrow$, and $\rightarrowtail$, which correspond to the ground, activating, and deactivating edges, respectively. The current evidence on whether an edge is active or inactive at any given moment is captured by the activation function $\mathcal{C} : E \rightarrow A^2$, while their internal details, as in Definition 5, are given by the functions $\dot{-}$ for ground edges and $\dot{\dot{-}}$ for hyper-edges.

The following definition will be used in the sequel to describe PRG whose values are totally consistent, totally inconsistent, or strictly consistent.

Definition 8. *Let* M *be an* (A, Σ)-PRG *and* $\mathcal{C} : E \rightarrow A^2$ *an activation function over the edges of* M*. We say that:*

- M *is* totally consistent *if for every edge* $x \in E$*, it holds that* $\mathcal{C}(x) \in \Delta_C$*.*
- M *is* totally inconsistent *if for every edge* $x \in E$*, it holds that* $\mathcal{C}(x) \in \Delta_P$*.*
- M *is* strictly consistent *if for every edge* $x \in E$*, it holds that* $\mathcal{C}(x) \in \Delta$*.*

Furthermore, since we are interested in consistency-preserving results, the following two theorems regarding Δ_C and Δ_P will be used in the subsequent subsections.

Theorem 1. Δ_C *is closed under the operators* $\neg$, $\sqcap\!\!\sqcap$, $\sqcup\!\!\sqcup$ *and* $\otimes$*.*

Proof. Let (tt_1, ff_1), $(tt_2, ff_2) \in \Delta_C$, then

- $\neg(\mathit{tt}_1, \mathit{ff}_1) = (\mathit{ff}_1, \mathit{tt}_1) \in \Delta_C$, that is, we intend to prove that
$$D((\mathit{ff}_1, \mathit{tt}_1), (0,0)) \leqslant D((\mathit{ff}_1, \mathit{tt}_1), (1,1)) \tag{5}$$
By hypothesis $(\mathit{tt}_1, \mathit{ff}_1) \in \Delta_C$, so it follows
$$\begin{aligned} & D((\mathit{tt}_1, \mathit{ff}_1), (0,0)) \leqslant D((\mathit{tt}_1, \mathit{ff}_1), (1,1)) \\ \Leftrightarrow & \sqrt{d(\mathit{tt}_1, 0)^2 + d(\mathit{ff}_1, 0)^2} \leqslant \sqrt{d(\mathit{tt}_1, 1)^2 + d(\mathit{ff}_1, 1)^2} \\ \Leftrightarrow & \sqrt{d(\mathit{ff}_1, 0)^2 + d(\mathit{tt}_1, 0)^2} \leqslant \sqrt{d(\mathit{ff}_1, 1)^2 + d(\mathit{tt}_1, 1)^2} \\ \Leftrightarrow & D((\mathit{ff}_1, \mathit{tt}_1), (0,0)) \leqslant D((\mathit{ff}_1, \mathit{tt}_1), (1,1)) \end{aligned}$$
Hence, if $(\mathit{tt}_1, \mathit{ff}_1) \in \Delta_C$, then $(\mathit{ff}_1, \mathit{tt}_1) \in \Delta_C$.

- $(\mathit{tt}_1, \mathit{ff}_1) \sqcap (\mathit{tt}_2, \mathit{ff}_2) = (\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcup \mathit{ff}_2) \in \Delta_C$, we intend to prove that
$$D((\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcup \mathit{ff}_2), (0,0)) \leqslant D((\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcup \mathit{ff}_2), (1,1)) \tag{6}$$
If $(\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcup \mathit{ff}_2)$ equals either $(\mathit{tt}_1, \mathit{ff}_1)$ or $(\mathit{tt}_2, \mathit{ff}_2)$, then (6) follows immediately from the hypothesis. If $(\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcup \mathit{ff}_2) = (\mathit{tt}_1, \mathit{ff}_2)$. By hypothesis,
$$\begin{aligned} & D((\mathit{tt}_2, \mathit{ff}_2), (0,0)) \leqslant D((\mathit{tt}_2, \mathit{ff}_2), (1,1)) \\ \Leftrightarrow & \sqrt{d(\mathit{tt}_2, 0)^2 + d(\mathit{ff}_2, 0)^2} \leqslant \sqrt{d(\mathit{tt}', 1)^2 + d(\mathit{ff}', 1)^2} \\ \Leftrightarrow & d(\mathit{tt}_2, 0)^2 + d(\mathit{ff}_2, 0)^2 \leqslant d(\mathit{tt}_2, 1)^2 + d(\mathit{ff}_2, 1)^2 \end{aligned} \tag{7}$$
Since $\mathit{tt}_1 \sqcap \mathit{tt}_2 = \mathit{tt}_1$ and $\mathit{ff}_1 \sqcup \mathit{ff}_2 = \mathit{ff}_2$ it follows $\mathit{tt}_1 \leqslant \mathit{tt}_2$ and $\mathit{ff}_1 \leqslant \mathit{ff}_2$. Thus, $d(\mathit{tt}_1, 0) \leqslant d(\mathit{tt}_2, 0)$ and $d(\mathit{tt}_1, 1) \geqslant d(\mathit{tt}_2, 1)$. Resorting again to (7), it follows
$$\begin{aligned} D((\mathit{tt}_1, \mathit{ff}_2), (0,0)) &= d(\mathit{tt}_1, 0)^2 + d(\mathit{ff}_2, 0)^2 \\ &\leqslant d(\mathit{tt}_2, 0)^2 + d(\mathit{ff}_2, 0)^2 \\ &\leqslant d(\mathit{tt}_2, 1)^2 + d(\mathit{ff}_2, 1)^2 \\ &\leqslant d(\mathit{tt}_1, 1)^2 + d(\mathit{ff}_2, 1)^2 = D((\mathit{tt}_1, \mathit{ff}_2), (1,1)) \end{aligned}$$
Similarly, it is possible to prove the case $(\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcup \mathit{ff}_2) = (\mathit{tt}_2, \mathit{ff}_1)$.

- $(\mathit{tt}_1, \mathit{ff}_1) \sqcup (\mathit{tt}_2, \mathit{ff}_2) \in \Delta_C$. Since $(\mathit{tt}_1, \mathit{ff}_1) \sqcup (\mathit{tt}_2, \mathit{ff}_2) = \neg(\neg(\mathit{tt}_1, \mathit{ff}_1) \sqcap \neg(\mathit{tt}_2, \mathit{ff}_2))$ it follows from the fact that Δ_C is closed under the operators $\neg$ and $\sqcap$.

- $(\mathit{tt}_1, \mathit{ff}_1) \otimes (\mathit{tt}_2, \mathit{ff}_2) = (\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcap \mathit{ff}_2) \in \Delta_C$. If $(\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcap \mathit{ff}_2)$ equals either $(\mathit{tt}_1, \mathit{ff}_1)$ or $(\mathit{tt}_2, \mathit{ff}_2)$, then it follows immediately from the hypothesis. If $(\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcap \mathit{ff}_2) = (\mathit{tt}_1, \mathit{ff}_2)$. By hypothesis,
$$\begin{aligned} & D((\mathit{tt}_1, \mathit{ff}_1), (0,0)) \leqslant D((\mathit{tt}_1, \mathit{ff}_1), (1,1)) \\ \Leftrightarrow & d(\mathit{tt}_1, 0)^2 + d(\mathit{ff}_1, 0)^2 \leqslant d(\mathit{tt}_1, 1)^2 + d(\mathit{ff}_1, 1)^2 \end{aligned} \tag{8}$$

Since $\mathit{tt}_1 \sqcap \mathit{tt}_2 = \mathit{tt}_1$ and $\mathit{ff}_1 \sqcap \mathit{ff}_2 = \mathit{ff}_2$ it follows $d(\mathit{ff}_1, 0) \geqslant d(\mathit{ff}_2, 0)$ and $d(\mathit{ff}_1, 1) \leqslant d(\mathit{ff}_2, 1)$. By also using (8), it follows

$$\begin{aligned} D((\mathit{tt}_1, \mathit{ff}_2), (0,0)) &= d(\mathit{tt}_1, 0)^2 + d(\mathit{ff}_2, 0)^2 \\ &\leqslant d(\mathit{tt}_1, 0)^2 + d(\mathit{ff}_1, 0)^2 \\ &\leqslant d(\mathit{tt}_1, 1)^2 + d(\mathit{ff}_1, 1)^2 \\ &\leqslant d(\mathit{tt}_1, 1)^2 + d(\mathit{ff}_2, 1)^2 = D((\mathit{tt}_1, \mathit{ff}_2), (1,1)) \end{aligned}$$

which proves $D((\mathit{tt}_1, \mathit{ff}_2), (0,0)) \leqslant D((\mathit{tt}_1, \mathit{ff}_2), (1,1))$. Similarly, it is possible to prove the case $(\mathit{tt}_1 \sqcap \mathit{tt}_2, \mathit{ff}_1 \sqcap \mathit{ff}_2) = (\mathit{tt}_2, \mathit{ff}_1)$.

□

It is worth noting that Δ_C is not closed under the operator $\oplus$.

For example, let $\mathbf{G}$ be the underlying lattice. Then, $(0.6, 0.3), (0.3, 0.6) \in \Delta_C$ however, $(0.6, 0.3) \oplus (0.3, 0.6) = (\max\{0.6, 0.3\}, \max\{0.3, 0.6\}) = (0.6, 0.6) \notin \Delta_C$.

The proof of the next Theorem is omitted, as it is similar to that of Theorem 1.

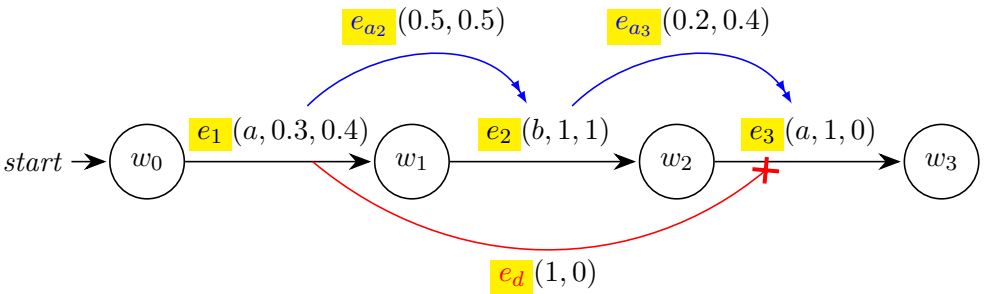
Theorem 2. *Δ_P is closed under the operators $\neg$, $\sqcap\!\!\sqcap$, $\sqcup\!\!\sqcup$ and $\oplus$.*

Δ_P is not closed under the operator $\otimes$.

For example, let $\mathbf{G}$ be the underlying lattice. Then, $(0.9, 0.2), (0.2, 0.9) \in \Delta_P$ however, $(0.9, 0.2) \otimes (0.2, 0.9) = (\min\{0.9, 0.2\}, \min\{0.2, 0.9\}) = (0.2, 0.2) \notin \Delta_P$.

The following example illustrates a PRG.

Example 10. *The following diagram represents a* $([0,1], \{a, b\})$-PRG.



Formally, this PRG *is defined over the set of states* $W = \{w_0, w_1, w_2, w_3\}$*, with initial state* w_0*. The set of ground edges is* $\rightarrow = \{e_1, e_2, e_3\}$*, the set of activating hyper-edges is* $\twoheadrightarrow = \{e_{a_2}, e_{a_3}\}$*, and the set of deactivating hyper-edges is* $\rightarrowtail = \{e_d\}$*.*

Each edge $e \in E$ is assigned a pair of truth values by the initial activation function $\mathcal{C}_0 : E \longrightarrow A^2$, as depicted in the diagram.

The functions $\overline{\cdot}$ and $\overline{\overline{\cdot}}$ map ground and hyper-edges, respectively, to their behavior. For example, $\overline{e_1} = (w_0, a, w_1)$ corresponds to the transition associated with ground edge e_1, and $\overline{\overline{e_d}} = (e_1, e_3)$ defines the deactivating hyper-edge e_d that deactivates e_3 upon the execution of e_1.

The following sections address non-conflicting and conflicting hyper-edges, respectively. Before formally defining these notions, we revisit the concepts defined in (1) and (2) with a few changes to accommodate the fact that edges are weighted by pairs of truth values in A^2. This revision is essential to enable a precise definition of what constitutes conflicting versus non-conflicting hyper-edges.

Definition 9. *Let M be an (A, Σ)-PRG where $E = (\rightarrow \cup \twoheadrightarrow \cup \rightarrowtail)$ is the set all of edges and $H = (\twoheadrightarrow \cup \rightarrowtail)$ is the set of all hyper-edges. For any $e_s, e_t \in E$ we define*

$$\begin{aligned} \mathsf{from}(e_s) &= \{h \in H \mid \exists e_t \in E \ \cdot \ \overline{\overline{h}} = (e_s, e_t)\} \\ \mathsf{to}(e_t) &= \{h \in H \mid \exists e_s \in E \ \cdot \ \overline{\overline{h}} = (e_s, e_t)\} \end{aligned}$$

Intuitively, as before, $\mathsf{from}(e_s)$ returns the set of hyper-edges that directly originate at edge e_s. Similarly, $\mathsf{to}(e_t)$ retrieves all hyper-edges that target edge e_t. We can now inductively define $\mathsf{from}^\star(e_s)$ as the transitive closure of $\mathsf{from}(e_s)$, which returns all hyper-edges that are (directly or indirectly) triggered by traversing e_s.

Definition 10. *Let M be an (A, Σ)-PRG. For any edge $e_s \in E$ we define inductively the set of all hyper-edges triggered by e_s as*

$$\mathsf{from}^\star(e_s) = \bigcup_{n \geqslant 0} \mathsf{from}^{(n)}(e_s)$$

where $\mathsf{from}^{(0)}(e_s) = \mathsf{from}(e_s)$ and $\mathsf{from}^{(n)}(e_s) = \bigcup_{h \in \mathsf{from}^{(n-1)}(e_s)} \mathsf{from}(h)$.

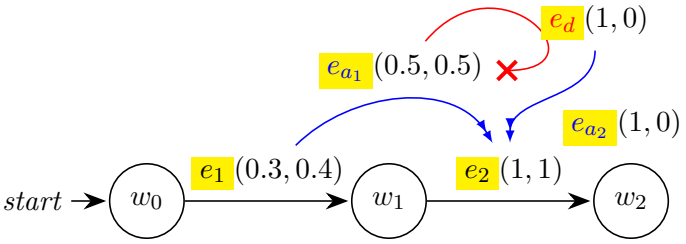
As before, we will define the sets $\mathsf{on}(e_s)$ (resp. $\mathsf{off}(e_s)$) as the set of edges $e_t \in E$ that are targeted by activating (resp. deactivating) hyper-edges $h \in H$ triggered by traversing e_s, i.e., $h \in \mathsf{from}^\star(e_s)$. Formally, for a PRG

Definition 11. *Let M be an (A, Σ)-PRG and $e_s \in E$ an edge. The set of edges activated (respectively deactivated) by e_s, written $\mathsf{on}(e_s)$ (respectively, $\mathsf{off}(e_s)$) are defined as follows,*

$$\begin{aligned} \mathsf{on}(e_s) &= \{e_t \in E \mid \exists h \in H \,.\, h \in (\mathsf{from}^\star(e_s) \cap \twoheadrightarrow \cap\, to(e_t))\} \\ \mathsf{off}(e_s) &= \{e_t \in E \mid \exists h \in H \,.\, h \in (\mathsf{from}^\star(e_s) \cap \rightarrowtail \cap\, to(e_t))\} \end{aligned} \tag{9}$$

The following example illustrates the definitions introduced above for PRG.

Example 11. *Consider the* $([0,1],\emptyset)$-PRG *depicted below.*[3] *Observe that edge* e_1 *triggers the activating hyper-edge* e_{a_1}*, which in turn triggers the deactivating hyper-edge* e_d*, which itself then triggers the activating hyper-edge* e_{a_2}*.*



Since $\mathsf{from}(e_1) = \{e_{a_1}\}$, $\mathsf{from}(e_{a_1}) = \{e_d\}$, *and* $\mathsf{from}(e_d) = \{e_{a_2}\}$ *it follows that* $\mathsf{from}^\star(e_1) = \{e_{a_1}, e_{a_2}, e_d\}$. *Next, since* $\mathsf{to}(e_2) = \{e_{a_1}, e_{a_2}\}$ *and* $\mathsf{to}(e_{a_1}) = \{e_d\}$ *one has* $\mathsf{on}(e_1) = \{e_2\}$ *and* $\mathsf{off}(e_1) = \{e_{a_1}\}$. *In other words, as expected, traversing* e_1 *ultimately triggers the hyper-edges* e_{a_1}*,* e_d*, and* e_{a_2}*, which in turn act upon* e_2 *and* e_{a_1} *as targets by activating and deactivating them, respectively.*

Finally, we introduce the notion of a set of hyper-edges adjacent to a pair of edges e_s and e_t.

Definition 12. *Let* M *be an* (A,Σ)-PRG *where* $E = (\rightarrow \cup \twoheadrightarrow \cup \not\rightarrow)$ *is the set all of edges. For any* $e_s, e_t \in E$ *the set* $\Gamma_A(e_s,e_t)$ *(resp.* $\Gamma_D(e_1,e_2)$*) denotes the set of activating (resp. deactivating) hyper-edges that are triggered by traversing edge* e_s *and that target edge* e_t*.*

$$\begin{aligned} \Gamma_A(e_s,e_t) &= \mathsf{from}^\star(e_s) \cap \mathsf{to}(e_t) \cap \twoheadrightarrow \\ \Gamma_D(e_s,e_t) &= \mathsf{from}^\star(e_s) \cap \mathsf{to}(e_t) \cap \not\rightarrow \end{aligned} \tag{10}$$

We write $\Gamma(e_s,e_t) = \Gamma_A(e_s,e_t) \cup \Gamma(e_s,e_t)$ *to denote the set of all hyper-edges that are triggered by traversing* e_s *and that target* e_t*.*

Note that if $h \in \Gamma_A(e_s,e_t)$, then $h \in \mathsf{to}(e_t)$. However, it does not necessarily follow that $h \in \mathsf{from}(e_s)$, since h may be triggered by the traversal of e_s but not directly originate from it (see hyper-edges e_{a_2} and e_d in Example 11). We use the

[3]Note that since the set of action symbols is empty, we omit them from the labels of the ground edges.

projection function $\pi_1 : E \times E \longrightarrow E$, defined by $\pi_1(e_1, e_2) = e_1$, and write $\pi_1(\overline{\overline{h}})$ to denote the edge from which the hyper-edge h originates.

Definition 12 plays a crucial role in the following sections. First, they allow us to formally define when a paraconsistent reactive graph contains no conflicting hyper-edges (see Definition 13). Second, they are fundamental for describing how the system's configuration evolves after traversing an edge.

When an edge e_s is traversed, it will trigger the activation of edges in $\mathsf{on}(e_s)$ and the deactivation of edges in $\mathsf{off}(e_s)$, based on the corresponding hyper-edges in $\Gamma(e_s, e_t)$. Therefore, updating the value of the target edges requires gathering information about the source edge, the hyper-edges it triggers, and the corresponding targets. This data must then be meaningfully combined to update the values of the target edges.

We have now introduced all the necessary tools for the following sections, which focus on three main tasks: formally defining conflicts between hyper-edges, describing how to update the system's configuration when an edge is traversed, and proposing policies for managing conflicts in the presence of contradictory hyper-edges.

4.1 Paraconsistent Reactive Graphs without Conflicts

In this section, we study paraconsistent reactive graphs without conflicting hyper-edges. Intuitively, this means that for any pair of edges, there is at most one hyper-edge connecting them. For instance, the paraconsistent reactive graph in Example 10 contains no conflicting hyper-edges, whereas the one in Example 11 includes a conflict between the activating hyper-edges e_{a_1} and e_{a_2}. Formally,

Definition 13. *Let M be an (A, Σ)-PRG. M is said to have no conflicting hyper-edges if for any $e_s, e_t \in E$, $|\Gamma(e_s, e_t)| \leqslant 1$.*

Intuitively, for any pair of edges e_s, e_t, the set of all hyper-edges triggered by e_s and targeting e_t, denoted by $\Gamma(e_s, e_t)$ as defined in Definition 12, contains at most one element.

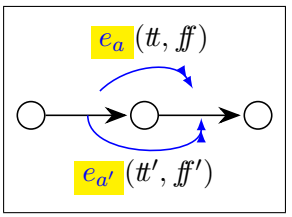


Figure 3

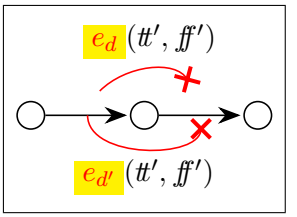


Figure 4

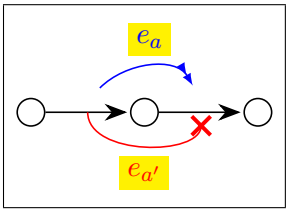


Figure 5

Visually, if $(tt, ff) \neq (tt', ff')$ are truth values in A^2, the scenarios depicted in Figure 3 and Figure 4 are not considered in this section, neither it is the scenario in Figure 5, regardless of the truth values of the edges[4].

It is important to note that in reactive graphs with bivalent hyper-edges (such as those in [11, 30, 42], or the $\mathrm{RG_P}$ introduced in Section 3) conflicting scenarios like those depicted in Figures 3 and 4 do not occur. In such systems, hyper-edges act as deterministic, bivalent switches. Therefore, having two distinct hyper-edges of the same type between the same source and target is redundant.

In contrast, the framework discussed in this paper allows hyper-edges to represent degrees of belief both in favor of and against their effect. As a result, even if two activating (or two deactivating) hyper-edges connect the same source and target, a conflict may arise if their values differ. This raises a natural question: which hyper-edge should take precedence, or how should their effects be combined?

In this section, however, we focus on PRG without conflicts. Our goal is to better understand how individual activating or deactivating hyper-edges influence the system's evolution. The main challenge lies in the fact that hyper-edges are weighted: their effect on a target edge depends not only on their own value, but also on the values of the edge that triggered them and the target edge itself.

More precisely, when an edge e_s is traversed and triggers a hyper-edge e_h targeting an edge e_t, it is necessary to define how to combine the values of e_s, e_h, and e_t to compute the updated value of e_t.

For this purpose, we now introduce two functions, Act and Deact. These functions are applied separately to activating and deactivating hyper-edges, and define how their individual effects impact the system. Note that these are not the only possible choices, but they are particularly simple.

Example 12. *The following functions,* Act *and* Deact*, combine truth values* α*,* β *and* γ *in* A^2 *into a single value in* A^2*.*

$$\mathsf{Act}(\alpha, \beta, \gamma) \;=\; (\alpha \sqcap \beta) \sqcup \gamma \qquad\qquad \mathsf{Deact}(\alpha, \beta, \gamma) \;=\; \neg(\alpha \sqcap \beta) \sqcap \gamma$$

Let $\alpha \in A^2$ *be the value of a traversed edge* e_s*, let* $\beta_a, \beta_d \in A^2$ *be the values of the triggered activating and deactivating hyper-edges* e_a *and* e_d*, respectively, and let* $\gamma \in A^2$ *denote the current value of a target edge* e_t*. There exists two possible scenarios which correspond to the activation of edge* e_t*, as depicted in Figure 6, or the deactivation of edge* e_t*, as depicted in Figure 7.*

The functions above are defined in order to generalize the expected behavior in the classical bivalent setting, where truth values are either $(1, 0)$ *(strictly true) or* $(0, 1)$ *(strictly false).*

[4]Note that the black edges depicted on Figures 3, 4 and 5 are not necessarily ground edges.

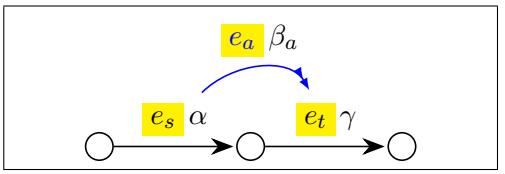


Figure 6

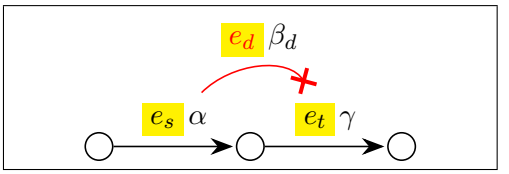


Figure 7

For the activation scenario depicted in Figure 6: when should the target edge e_t be activated? There are two cases to consider:

> ***Case 1:*** *e_t is already active. In this case, any attempt to activate e_t has no effect. Formally, component $\ldots \sqcup \gamma$ in function* Act *ensures that the returned value is at least as great as γ.*
>
> ***Case 2:*** *The traversed edge e_s triggers an activating hyper-edge e_a. In this case, the values α and β_a, associated to e_s and e_a, respectively, reinforce the activation of e_t. This is captured by the term $\alpha \sqcap \beta_a$ in function* Act.

Similarly, for the deactivation scenario in Figure 7: when should the target edge e_t be deactivated? Again, there are two scenarios to consider:

> ***Case 1:*** *e_t is already deactivated. In this case, any attempt to deactivate it again has no effect. Formally, component $\ldots \sqcap \gamma$ in function* Deact *ensures that the returned value is at best equal to γ.*
>
> ***Case 2:*** *The traversed edge e_s triggers a deactivating hyper-edge e_d. In this case, the values α and β_d, associated to e_s and e_d, respectively, negatively reinforce edge e_t. This is captured by $\neg(\alpha \sqcap \beta_d)$ in function* Deact.

These cases extend naturally to general values in A^2 via the twist-structure operators employed. As discussed in the sequel, functions Act *and* Deact *define how the values of the traversed edge, the triggered hyper-edge, and the target edge are combined to compute the new value of e_t.*

It is important to highlight that the functions Act and Deact are defined to generalize classic bivalent behavior. That is, if all edges are weighted by the pair $(1, 0)$, that represents absolute truth, or pair $(0, 1)$, that represents absolute falsity, then the output of these functions coincides with the expected result from classical reactive graphs. This important property is formalized in Proposition 1.

It is also important to note that similar challenges have also been addressed in the research on fuzzy reactive graphs [7, 8, 9], where all edges are assigned weights in

the interval $[0, 1]$. In that setting, the authors use aggregation functions to combine values in $[0, 1]$. Formally, an aggregation function is a mapping $\mathsf{A}^{(n)} : [0, 1]^n \to [0, 1]$ that is monotonic in each variable and satisfies certain boundary conditions [32]. More recent work extended this concept to any bounded poset [21, 36].

However, when edges in a fuzzy reactive graph are restricted to classical truth values, i.e., transitions are either fully possible (1) or not (0)– the output of some aggregation functions do not coincide with the expected bivalent outcomes. For this reason, we refrain from adopting aggregation functions in here to ensure that our framework, when restricted to classical truth values, aligns with the classic theory on reactive graphs.

The following proposition ensures that the functions Act and Deact correspond to the classical activation and deactivation of edges, respectively, when the input truth values are classical, i.e., $(1, 0)$ and $(0, 1)$. In this setting, if all edges are weighted by the pair $(1, 0)$, providing evidence in favour of traversing the edge, or by the pair $(0, 1)$, providing the opposite, then the output of these functions coincides with the expected behaviour of classical reactive graphs.

More specifically, as noted in Example 12, an activating edge e_a triggered by a ground edge e_s and targeting an edge e_t, as depicted in Figure 6, results in the activation of e_t either when e_t is already active or when both e_a and e_s are active. Likewise, a deactivating edge e_d triggered by a ground edge e_s and targeting an edge e_t, as shown in Figure 7, results in the deactivation of e_t either when e_t is already inactive or when both e_d and e_s are active. Formally,

Proposition 1. *Let M be an (A, Σ)-PRG with no conflicting hyper-edges such that, for any edge $x \in E$, $\mathcal{C}(x) \in \{(1, 0), (0, 1)\}$. For any ground edges $e_s, e_t \in \rightarrow$, activating hyper-edge $e_a \in \twoheadrightarrow$, and deactivating hyper-edge $e_d \in \rightarrowtail$, it follows that:*

$$\mathsf{Act}(\mathcal{C}(e_s), \mathcal{C}(e_a), \mathcal{C}(e_t)) = \begin{cases} (1,0) & \text{if } \mathcal{C}(e_t) = (1,0) \text{ or } \mathcal{C}(e_s) = \mathcal{C}(e_a) = (1,0) \\ (0,1) & \text{otherwise} \end{cases}$$

$$\mathsf{Deact}(\mathcal{C}(e_s), \mathcal{C}(e_d), \mathcal{C}(e_t)) = \begin{cases} (0,1) & \text{if } \mathcal{C}(e_t) = (0,1) \text{ or } \mathcal{C}(e_s) = \mathcal{C}(e_a) = (1,0) \\ (1,0) & \text{otherwise} \end{cases}$$

Proof. Let M be an (A, Σ)-PRG with no conflicting hyper-edges. Consider two edges $e_s, e_t \in E$, and let $e_a \in \twoheadrightarrow$ and $e_d \in \rightarrowtail$ be an activating and a deactivating hyper-edge, respectively. Let α denote the value of the traversed edge e_s, β_a and β_d the values of the triggered hyper-edges e_a and e_d, and γ the current value of the target edge e_t. We consider the two scenarios illustrated in Figures 6 and 7.

If values α, β_a, β_d, and γ are restricted to classical truth values, i.e., $(1,0)$ (true) and $(0,1)$ (false), then the behavior of functions Act and Deact coincides with the classical theory.

We first consider the outputs of $\mathsf{Act}(\alpha, \beta_a, \gamma)$, which give the updated truth value of e_t after traversing e_s in the activation scenario of Figure 6.

If $\alpha = (1,0)$, then

$\mathsf{Act}(\alpha, \beta_a, \gamma)$	$\gamma = (0,1)$	$\gamma = (1,0)$
$\beta_a = (0,1)$	$(0,1)$	$(1,0)$
$\beta_a = (1,0)$	$(1,0)$	$(1,0)$

If $\alpha = (0,1)$, then

$\mathsf{Act}(\alpha, \beta_a, \gamma)$	$\gamma = (0,1)$	$\gamma = (1,0)$
$\beta_a = (0,1)$	$(0,1)$	$(1,0)$
$\beta_a = (1,0)$	$(0,1)$	$(1,0)$

As expected, the edge e_t is activated if and only if either it was already active, i.e., $\gamma = (1,0)$, or if both the traversed edge and the activating hyper-edge are active, i.e., $\alpha = \beta_a = (1,0)$.

Next, we consider $\mathsf{Deact}(\alpha, \beta_d, \gamma)$, which computes the updated value of e_t in the deactivation scenario depicted in Figure 7.

If $\alpha = (1,0)$, then

$\mathsf{Deact}(\alpha, \beta_d, \gamma)$	$\gamma = (0,1)$	$\gamma = (1,0)$
$\beta_d = (0,1)$	$(0,1)$	$(1,0)$
$\beta_d = (1,0)$	$(0,1)$	$(0,1)$

If $\alpha = (0,1)$, then

$\mathsf{Deact}(\alpha, \beta_d, \gamma)$	$\gamma = (0,1)$	$\gamma = (1,0)$
$\beta_d = (0,1)$	$(0,1)$	$(1,0)$
$\beta_d = (1,0)$	$(0,1)$	$(1,0)$

Again, as expected, the edge e_t is deactivated if and only if either it was already inactive, i.e., $\gamma = (0,1)$, or if both the traversed edge and the deactivating hyper-edge are active, i.e., $\alpha = \beta_d = (1,0)$. □

The following proposition studies monotonicity properties of functions Act and Deact introduced in Example 12.

Proposition 2. *Let $\alpha, \alpha', \beta, \beta', \gamma$ and γ' be in A^2. Then the following conditions hold with respect to the truth-order $\leqslant_t$.*

1. *If $\alpha \leqslant_t \alpha'$, $\beta \leqslant_t \beta'$ and $\gamma \leqslant_t \gamma'$, then $\mathsf{Act}(\alpha, \beta, \gamma) \leqslant_t \mathsf{Act}(\alpha', \beta', \gamma')$.*
2. *If $\alpha \leqslant_t \alpha'$, $\beta \leqslant_t \beta'$ and $\gamma \geqslant_t \gamma'$, then $\mathsf{Deact}(\alpha, \beta, \gamma) \geqslant_t \mathsf{Deact}(\alpha', \beta', \gamma')$.*

And the following conditions hold with respect to the knowledge-order $\leqslant_k$.

3. *If $\alpha \leqslant_k \alpha'$, $\beta \leqslant_k \beta'$ and $\gamma \leqslant_k \gamma'$, then $\mathsf{Act}(\alpha, \beta, \gamma) \leqslant_k \mathsf{Act}(\alpha', \beta', \gamma')$.*
4. *If $\alpha \leqslant_k \alpha'$, $\beta \leqslant_k \beta'$ and $\gamma \leqslant_k \gamma'$, then $\mathsf{Deact}(\alpha, \beta, \gamma) \leqslant_k \mathsf{Deact}(\alpha', \beta', \gamma')$.*

Proof. The proof relies on well-known properties of twist-structures and the associated orders $\leqslant_t$ and $\leqslant_k$. In particular, we use the fact that De Morgan laws hold, e.g., $\neg(\alpha \sqcap \beta) = \neg\alpha \sqcup \neg\beta$ (see [19, Lemma 2]). Also, since twist-structures form (square) bilattices (see [1, Definition 2.1]), it follows that:

$$\text{if } \alpha \leqslant_t \alpha', \text{ then } \neg\alpha \geqslant_t \neg\alpha' \tag{11}$$

$$\text{if } \alpha \leqslant_k \alpha', \text{ then } \neg\alpha \leqslant_k \neg\alpha' \tag{12}$$

The conditions 1 and 3 follow immediately from the monotonicity of $\sqcap$ and $\sqcup$ in the underlying lattice $(A, \leqslant)$, which lifts to the monotonicity of $\barwedge$ and $\barvee$ with respect to $\leqslant_t$ and $\leqslant_k$.

To prove condition 2 we start with the hypothesis $\alpha \leqslant_t \alpha'$ and $\beta \leqslant_t \beta'$. It follows from (11) and the monotonicity of $\barvee$ that $\neg\alpha \barvee \neg\beta \geqslant_t \neg\alpha' \barvee \neg\beta'$. By De Morgan, this is equivalent to $\neg(\alpha \barwedge \beta) \geqslant_t \neg(\alpha' \barwedge \beta')$. Since $\gamma \geqslant_t \gamma'$ and $\barwedge$ is monotonic, we conclude $\neg(\alpha \barwedge \beta) \barwedge \gamma \geqslant_t \neg(\alpha' \barwedge \beta') \barwedge \gamma'$, that is, $\mathsf{Deact}(\alpha, \beta, \gamma) \geqslant_t \mathsf{Deact}(\alpha', \beta', \gamma')$.

Similarly, to prove 4 we start with the hypothesis $\alpha \leqslant_k \alpha'$ and $\beta \leqslant_k \beta'$. It follows from (12), the monotonicity of $\barvee$ and De Morgan laws that $\neg\alpha \barvee \neg\beta \leqslant_k \neg\alpha' \barvee \neg\beta'$. Finally, since $\gamma \leqslant_k \gamma'$ and by the monotonicity of $\barwedge$, we prove that $\mathsf{Deact}(\alpha, \beta, \gamma) \leqslant_k \mathsf{Deact}(\alpha', \beta', \gamma')$. □

Informally, condition 1 in the previous proposition states that the activation function, Act, is monotonic in the truth order: the stronger the evidence for the traversed edge, the activating hyper-edge, and the target edge, the stronger the activation effect. Condition 2 describes the expected behavior of the deactivation function, Deact, the higher the belief in the traversed and deactivating edges, the more the function impacts the target towards deactivation (i.e., a lower output), and the more active the target edge is initially, the less impact the deactivation will have. Conditions (3) and (4) confirm that both Act and Deact are monotonic with respect to the knowledge order ($\leqslant_k$), i.e., more information in the inputs leads to more information in the output.

As in Section 3, we now aim at defining a semantics for "unfolding" PRG with no conflicting hyper-edges into a paraconsistent graph. For this, we first clarify how the activation function of a PRG changes after traversing an edge, using the functions Act and Deact introduced in Definition 12.

Definition 14. *Given an (A, Σ)-paraconsistent reactive graph with non conflicting hyper-edges let $\mathcal{C} : E \to A^2$ be a activation function and $e, x \in E$, then*

$$\mathcal{C}^e(x) = \begin{cases} \mathcal{C}(x) & \text{if } x \notin (\mathsf{on}(e) \cup \mathsf{off}(e)) \\ \mathsf{Act}(\mathcal{C}(\pi_1(\overline{h})), \mathcal{C}(h), \mathcal{C}(x)) & \text{if } \exists h \in H \cdot h \in \Gamma_A(e, x) \\ \mathsf{Deact}(\mathcal{C}(\pi_1(\overline{h})), \mathcal{C}(h), \mathcal{C}(x)) & \text{if } \exists h \in H \cdot h \in \Gamma_D(e, x) \end{cases}$$

is the updated value of the edge x, i.e. $\mathcal{C}(x)$, after transvesing e.

In other words, if edge x is not affected by any hyper-edge triggered by e, its value remains unchanged. Otherwise, since we assume no conflicting hyper-edges, there exists a unique hyper-edge h triggered by e[5] that targets x. If h is activating (resp. deactivating), then the updated value of x is computed using Act (resp. Deact), based on the current values of $\pi_1(\overline{h})$, h, and x, given by the current activation function $\mathcal{C}$.

We are now equipped to define the semantics for PRG without conflicting hyper-edges. As before, we begin from an initial configuration $\langle w_0, \mathcal{C}_0 \rangle$. Then, for any ground edge $e \in \rightarrow$ going out of state w_0, we compute the updated activation function $\mathcal{C}^e(x)$ for all edges x according to Definition 14.

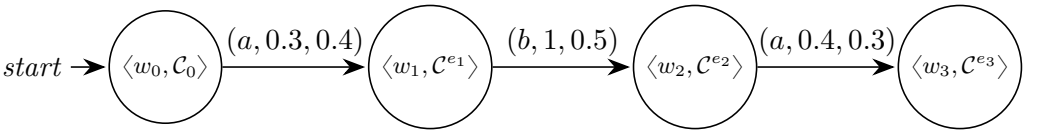
Definition 15. *The semantics of an (A, Σ)-paraconsistent graph P is given by evolving a configuration $\langle w, \mathcal{C} \rangle$, with $w \in W$ and $\mathcal{C} : E \to A^2$, starting from $\langle w_0, \mathcal{C}_0 \rangle$, as defined by the following rule.*

$$\frac{\exists e \in \rightarrow \ \cdot \ \overline{e} \ = \ (w, a, w')}{\langle w, \mathcal{C} \rangle \xrightarrow{a, \mathcal{C}(e)}_P \langle w', \mathcal{C}^e \rangle}$$

In practice, we are only interested in transversing ground edges $e \in \rightarrow$ such that $\mathcal{C}(e) \neq (0, 1)$, which corresponds to edges that are not inactive.

Let us apply the semantics defined above to the PRG in Example 10.

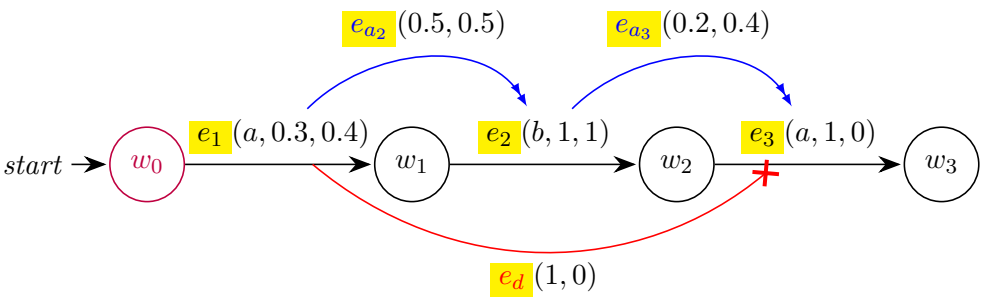
Example 13. *Consider the* $([0, 1], \{a, b\})$-PRG *without conflicting hyper-edges from Example 10, and the following* $([0, 1], \{a, b\})$*-paraconsistent graph P:*



We will show that unfolding the PRG *from Example 10, according to the semantics defined in Definition 15, yields the standard paraconsistent graph P.*

The initial configuration is $\langle w_0, \mathcal{C}_0 \rangle$, where the current state is w_0, colored purple, and $\mathcal{C}_0$ assigns to each edge in E the values as depicted below.

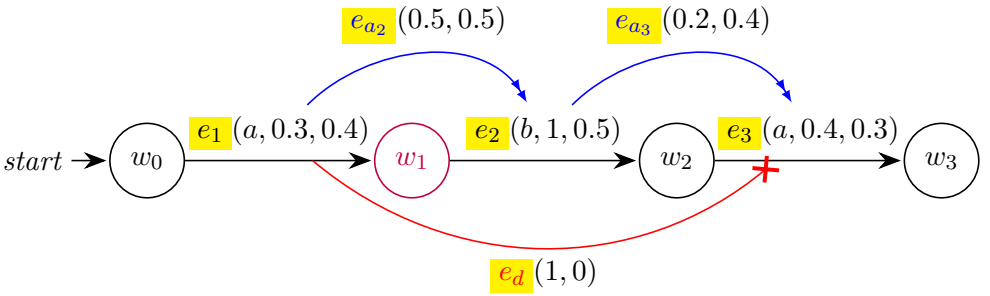
[5] Recall that $\pi_1(\overline{h})$ denotes the edge from which the hyper-edge h originates. While h is triggered by e_s, it does not necessarily originate directly from e_s.



There exists a ground edge e_1 *such that* $\overline{e_1} = (w_0, a, w_1)$. *We now compute for any* $x \in E$ *the updated activation function* $\mathcal{C}^{e_1}(x)$ *after traversing* e_1. *Since* $\mathsf{from}^\star(e_1) = \{e_{a_2}, e_d\}$ *and* $\mathsf{on}(e_1) \cup \mathsf{off}(e_1) = \{e_2, e_3\}$, *the updated activation function is:*

$$\mathcal{C}^{e_1}(x) = \begin{cases} \mathcal{C}(x) & \text{if } x \in E \backslash \{e_2, e_3\} \\ (1, 0.5) & \text{if } x = e_2 \\ (0.4, 0.3) & \text{if } x = e_3 \end{cases}$$

This corresponds to the transition $\langle w_0, \mathcal{C}_0 \rangle \xrightarrow{(a,0.3,0.4)}_P \langle w_1, \mathcal{C}^{e_1} \rangle$ *in the paraconsistent graph* P. *After transversing edge* e_1, *the updated* PRG *in 10 evolves to*



The system is now at state w_1, *colored purple, and there exists* e_2 *such that* $\overline{e_2} = (w_1, b, w_2)$. *Since* $\mathsf{from}^\star(e_2) = \{e_{a_3}\}$ *and* $\mathsf{on}(e_2) \cup \mathsf{off}(e_2) = \{e_3\}$, *the updated activation function is:*

$$\mathcal{C}^{e_2}(x) = \begin{cases} \mathcal{C}^{e_1}(x) & \text{if } x \in E \backslash \{e_3\} \\ (0.4, 0.3) & \text{if } x = e_3 \end{cases}$$

This results in the transition $\langle w_1, \mathcal{C}^{e_1} \rangle \xrightarrow{(1,0.5)}_P \langle w_2, \mathcal{C}^{e_2} \rangle$. *Finally, traversing* e_3 *leads to no further updates, since it triggers no hyper-edges, i.e.,* $\mathcal{C}^{e_3}(x) = \mathcal{C}^{e_2}(x)$

for any $x \in E$. Thus, this corresponds to the transition $\langle w_2, \mathcal{C}^{e_2} \rangle \xrightarrow{(0.4,0.3)}_P \langle w_3, \mathcal{C}^{e_3} \rangle$ in the paraconsistent graph P.

In summary, the dynamic behavior of the PRG *in Example 10 can be "unfolded" into the standard paraconsistent graph P. After unfolding, the effects of all hyper-edges have been incorporated into the ground edges, and the hyper-edges themselves may be omitted from the diagram.*

Finally, we conclude this section by showing that a paraconsistent reactive graph without conflicting hyper-edges maintains consistency throughout its evolution, as long as all initial ground and hyper-edges are not weighted by inconsistent truth values in A^2. More specifically, if M is a PRG without conflicting hyper-edges and it is totally consistent, then the system evolves without ever generating inconsistency.

Proposition 3. *Let M be a* PRG *without conflicting hyper-edges. If M is totally consistent, that is, $\mathcal{C}_0(x) \in \Delta_C$ for every edge $x \in E$, then the system evolves from the initial configuration by traversing ground edges $e_1, \ldots, e_n$ to subsequent configurations, as given by Definition 15, written as*

$$\langle w_0, C_0 \rangle \xrightarrow{a \in \Sigma, \mathcal{C}(e_1)}_P \langle w_1, C_1 \rangle \xrightarrow{a \in \Sigma, \mathcal{C}(e_2)}_P \cdots \xrightarrow{a \in \Sigma, \mathcal{C}(e_n)}_P \langle w_n, C_n \rangle.$$

and for every edge $x \in E$, it follows that $C_1(x), \ldots, C_n(x) \in \Delta_C$.

Proof. The proof follows directly from the closure of set Δ_C under the operations $\neg$, $\sqcap\!\!\sqcap$, and $\sqcup\!\!\sqcup$ (see Theorem 1). Since functions Act and Deact are defined using only these operators, they preserve consistency, that is, if their inputs are in Δ_C, then their outputs are also in Δ_C.

Now, consider the semantic rule in Definition 15. After traversing a ground edge e, any edge $x \in E$ its truth value is either:

- not affected by any hyper-edge triggered by e, i.e., $x \notin \mathsf{on}(e) \cup \mathsf{off}(e)$. In this case, the truth value $\mathcal{C}(x)$ remains unchanged and, by hypothesis, is already in Δ_C, or
- updated using Act or Deact functions, since the inputs are themselves in Δ_C, the updated value of $\mathcal{C}(x)$ is also in Δ_C.

□

4.2 Paraconsistent Reactive Graphs with Conflicts

In this section, we explore policies for managing conflicting hyper-edges in paraconsistent reactive graphs. Recall from Definition 13 that a paraconsistent reactive

graph M, as defined in Definition 7, has conflicting hyper-edges if there exist edges $e_s, e_t \in E$ such that $|\Gamma(e_s, e_t)| > 1$, that is, more than one hyper-edge is triggered by e_s and targets e_t.

In the previous section (Section 4.1), we restricted our attention to paraconsistent reactive graphs whose hyper-edges could be internally inconsistent but never conflicted with one another. Moreover, we showed in Proposition 3 that as long as the system starts with a consistent activation function, it evolves without ever becoming inconsistent.

We now lift this restriction to the general case in which multiple, potentially contradictory hyper-edges target the same edge. This allows us to address scenarios such as those depicted in Figures 3, 4 and 5. This challenge raises fundamental questions, such as: How should we combine the information provided by multiple hyper-edges? Should certain hyper-edges take precedence over others? If so, which policies should guide such choices of precedence?

In bivalent switch graphs resolving such conflicts is straightforward: one simply chooses to prioritize either activation or deactivation [42]. In this section, however, rather than relying solely on binary resolution policies, we consider more flexible approaches that allow multiple hyper-edges to contribute their beliefs. For this purpose, we introduce different functions that combine the information provided by all relevant hyper-edges into a single representative value. Each one is guided by a specific conflict resolution policy that motivates and justifies the chosen strategy.

Let us start by introducing functions $\mathsf{gather}^{\mathsf{A}}_{\odot}$ and $\mathsf{gather}^{\mathsf{D}}_{\odot}$ which will be used to combine the information provided by multiple conflicting activating and deactivating hyper-edges, respectively.

Definition 16. *Let M be any (A, Σ)-PRG where $E = (\rightarrow \cup \twoheadrightarrow \cup \rightarrowtail)$ is the set of all edges. For $\odot \in \{\sqcap\!\!\sqcap, \sqcup\!\!\sqcup\}$ the function*

$$\begin{aligned} \mathsf{gather}^{A}_{\odot} :\ & E \times E \longrightarrow A^2 \\ & e_s, e_t \longmapsto \bigodot_{h \in \Gamma_A(e_s, e_t)} \mathsf{Act}(\mathcal{C}(\pi_1(\overline{\overline{h}})), \mathcal{C}(h), \mathcal{C}(e_t)) \end{aligned}$$

computes the updated value of e_t by combining the information of all activating hyper-edges triggered by e_s and that target e_t and the function

$$\begin{aligned} \mathsf{gather}^{D}_{\odot} :\ & E \times E \longrightarrow A^2 \\ & e_s, e_t \longmapsto \bigodot_{h \in \Gamma_A(e_s, e_t)} \mathsf{Deact}(\mathcal{C}(\pi_1(\overline{\overline{h}})), \mathcal{C}(h), \mathcal{C}(e_t)) \end{aligned}$$

computes the updated value of e_t by combining the information of all deactivating hyper-edges triggered by e_s and that target e_t.

Intuitively, $\mathsf{gather}^A_{\sqcup\!\!\sqcup}$ and $\mathsf{gather}^D_{\sqcup\!\!\sqcup}$ correspond to a "best-case" scenario: the value of e_t is updated to the highest possible result among all contributions. Conversely, $\mathsf{gather}^A_{\sqcap\!\!\sqcap}$ and $\mathsf{gather}^D_{\sqcap\!\!\sqcap}$ represent a "worst-case" scenario, where the lowest result is chosen.

For example, if there are conflicting activating hyper-edges targeting e_t, then $\mathcal{C}(e_t)$ is updated to $(1,0)$ by $\mathsf{gather}^A_{\sqcup\!\!\sqcup}$ if at least one hyper-edge $h \in \Gamma_A(e_s, e_t)$ completely supports activation, i.e., $\mathsf{Act}(\mathcal{C}(\pi_1(\overline{h})), \mathcal{C}(h), \mathcal{C}(e_t)) = (1,0)$. On the other hand, $\mathsf{gather}^A_{\sqcap\!\!\sqcap}$ requires that all hyper-edges in $\Gamma_A(e_s, e_t)$ support complete activation of e_t.

Similarly, if there are conflicting deactivating hyper-edges targeting e_t, then e_t is completely deactivated by $\mathsf{gather}^D_{\sqcup\!\!\sqcup}$, i.e. $\mathcal{C}(e_t)$ is updated to $(0,1)$, only if every $h \in \Gamma_D(e_s, e_t)$ supports complete deactivation of e_t, i.e., $\mathsf{Deact}(\mathcal{C}(\pi_1(\overline{h})), \mathcal{C}(h), \mathcal{C}(e_t)) = (0,1)$. By contrast, $\mathsf{gather}^D_{\sqcap\!\!\sqcap}$ requires only one such hyper-edge to support complete deactivation of e_t.

Of course the choice of operator $\odot$ depends on the problem at hand. Informally, choosing $\sqcap\!\!\sqcap$ reflects:

- a conservative stance for activation, requiring consensus among hyper-edges;
- a pessimistic stance for deactivation, where limited evidence suffices to deactivate.

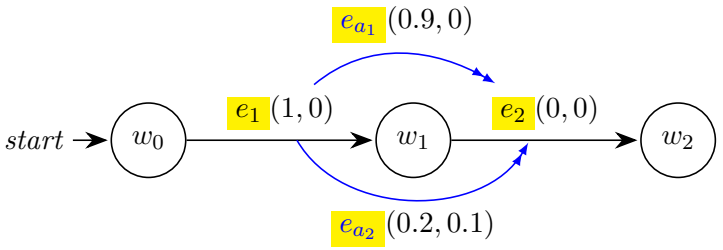
In contrast, using $\sqcup\!\!\sqcup$ corresponds to:

- a lenient stance for activation, where limited evidence is sufficient;
- an optimistic stance for deactivation, where full consensus is needed to justify deactivation.

Thus, if we face a conflict between hyper-edges of the same type, as in Figure 3 and Figure 4, we can use $\mathsf{gather}^A_{\odot}$ and $\mathsf{gather}^D_{\odot}$, respectively, to update the value of the target edge, depending on whether we value any available evidence or require consensus among all hyper-edges.

The following two examples illustrate the different conflict resolution policies defined above.

Example 14. *Consider the following* $([0,1], \emptyset)$-PRG *with conflicted hyper-edges.*



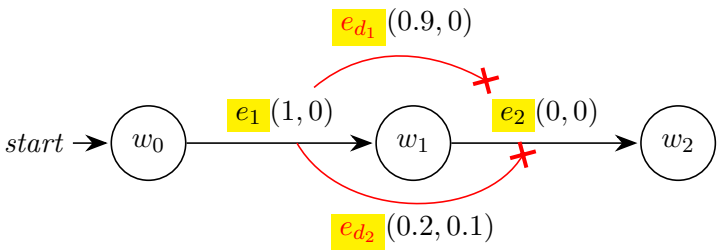
Consequently,

$$\mathsf{Act}(\mathcal{C}(e_1), \mathcal{C}(e_{a_1}), \mathcal{C}(e_2)) = ((1,0) \sqcap (0.9,0)) \sqcup (0,0) = (0.9, 0)$$
$$\mathsf{Act}(\mathcal{C}(e_1), \mathcal{C}(e_{a_2}), \mathcal{C}(e_2)) = ((1,0) \sqcap (0.2,0.1)) \sqcup (0,0) = (0.2, 0)$$

Since $(0.2, 0.1) \leqslant_t (0.9, 0)$*, i.e.* $\mathcal{C}(e_{a_2}) \leqslant_t \mathcal{C}(e_{a_1})$*, it follows from Proposition 2 it follows that* $\mathsf{Act}(\mathcal{C}(e_1), \mathcal{C}(e_{a_2}), \mathcal{C}(e_2)) \leqslant_t \mathsf{Act}(\mathcal{C}(e_1), \mathcal{C}(e_{a_1}), \mathcal{C}(e_2))$.

Additionally, we have $\mathsf{gather}^A_{\sqcap}(e_1, e_2) = (0.2, 0)$ *and* $\mathsf{gather}^A_{\sqcup}(e_1, e_2) = (0.9, 0)$ *which shows that using* $\sqcap$ *to combine the information from the activating hyper-edges results in a more conservative (weaker) activation of* e_2*, whereas using* $\sqcup$ *yields a more lenient (stronger) activation.*

Example 15. *Consider the following* $([0, 1], \emptyset) - \mathrm{PRG}$ *with conflicted hyper-edges.*



Since $\mathcal{C}(e_{d_2}) \leqslant_t \mathcal{C}(e_{d_1})$*, it follows from Proposition 2 that*

$$(0, 0.2) = \mathsf{Deact}(\mathcal{C}(e_1), \mathcal{C}(e_{d_2}), \mathcal{C}(e_2)) \geqslant_t \mathsf{Deact}(\mathcal{C}(e_1), \mathcal{C}(e_{d_1}), \mathcal{C}(e_2)) = (0, 0.9)$$

Intuitively, hyper-edge e_{d_1} *has a higher truth value than* e_{d_2} *so it produces a stronger deactivation effect on edge* e_2*.*

Additionally, $\mathsf{gather}^D_{\sqcap}(e_1, e_2) = (0, 0.9)$ *and* $\mathsf{gather}^D_{\sqcup}(e_1, e_2) = (0, 0.2)$*, which illustrates that using the operator* $\sqcap$ *results in a more pessimistic stance, pushing the value closer to strictly false* $(0, 1)$*, while* $\sqcup$ *leads to a more optimistic outcome, with a weaker deactivation.*

The following proposition provides a simplification in the calculation of $\mathsf{gather}^A_{\odot}$ and $\mathsf{gather}^D_{\odot}$ if all hyper-edges in $\Gamma_A(e_s, e_t)$ or $\Gamma_D(e_s, e_t)$ originate directly from e_s. That is, no hyper-edge triggered by e_s triggers another hyper-edge that also targets e_t. Formally, $\pi_1(\overline{h}) = e_s$ for every $h \in \Gamma_A(e_s, e_t)$ or $h \in \Gamma_D(e_s, e_t)$.

Proposition 4. *Let M be an (A, Σ)-PRG. For $e_s, e_t \in E$, such that $\Gamma_A(e_s, e_t) \neq \emptyset$, if $\pi_1(\overline{h}) = e_s$ for every $h \in \Gamma_A(e_s, e_t)$, then*

$$\bigsqcup_{h \in \Gamma_A(e_s,e_t)} \mathsf{Act}(\mathcal{C}(e_s), \mathcal{C}(h), \mathcal{C}(e_t)) = \mathsf{Act}\left(\mathcal{C}(e_s), \bigsqcup_{h \in \Gamma_A(e_s,e_t)} \mathcal{C}(h), \mathcal{C}(e_t)\right) \tag{13}$$

$$\bigsqcap_{h \in \Gamma_A(e_s,e_t)} \mathsf{Act}(\mathcal{C}(e_s), \mathcal{C}(h), \mathcal{C}(e_t)) = \mathsf{Act}\left(\mathcal{C}(e_s), \bigsqcap_{h \in \Gamma_A(e_s,e_t)} \mathcal{C}(h), \mathcal{C}(e_t)\right) \tag{14}$$

For $e_s, e_t \in E$, such that $\Gamma_D(e_s, e_t) \neq \emptyset$, if $\pi_1(\overline{h}) = e_s$ for every $h \in \Gamma_D(e_s, e_t)$, then

$$\bigsqcap_{h \in \Gamma_D(e_s,e_t)} \mathsf{Deact}(\mathcal{C}(e_s), \mathcal{C}(h), \mathcal{C}(e_t)) = \mathsf{Deact}\left(\mathcal{C}(e_s), \bigsqcup_{h \in \Gamma_D(e_s,e_t)} \mathcal{C}(h), \mathcal{C}(e_t)\right) \tag{15}$$

$$\bigsqcup_{h \in \Gamma_D(e_s,e_t)} \mathsf{Deact}(\mathcal{C}(e_s), \mathcal{C}(h), \mathcal{C}(e_t)) = \mathsf{Deact}\left(\mathcal{C}(e_s), \bigsqcap_{h \in \Gamma_D(e_s,e_t)} \mathcal{C}(h), \mathcal{C}(e_t)\right) \tag{16}$$

Proof. We prove (13) and (15). Results (14) and (16) follow similarly. For some $m, n \in \mathbb{N}$, let $\Gamma_A(e_s, e_t) = \{h^A_1, h^A_2, \ldots, h^A_m\}$ be the set of activating hyper-edges that are triggered by e_s and target e_t and $\Gamma_D(e_s, e_t) = \{h^D_1, h^D_2, \ldots, h^D_n\}$ the set of deactivating hyper-edges that are triggered by e_s and target e_t. The proof of (13) is as follows

$$\begin{aligned}
&\bigsqcup_{h \in \Gamma_A(e_s,e_t)} \mathsf{Act}(\mathcal{C}(e_s), \mathcal{C}(h), \mathcal{C}(e_t)) \\
&= \mathsf{Act}(\mathcal{C}(e_s), \mathcal{C}(h^A_1), \mathcal{C}(e_t)) \sqcup \ldots \sqcup \mathsf{Act}(\mathcal{C}(e_s), \mathcal{C}(h^A_m), \mathcal{C}(e_t)) && \{\text{defn. of } \mathsf{Act}\} \\
&= ((\mathcal{C}(e_s) \sqcap \mathcal{C}(h^A_1)) \sqcup \mathcal{C}(e_t)) \sqcup \ldots \sqcup ((\mathcal{C}(e_s) \sqcap \mathcal{C}(h^A_m)) \sqcup \mathcal{C}(e_t)) && \{\sqcup \text{ is involutive}\} \\
&= (\mathcal{C}(e_s) \sqcap \mathcal{C}(h^A_1)) \sqcup \ldots \sqcup (\mathcal{C}(e_s) \sqcap \mathcal{C}(h^A_m)) \sqcup \mathcal{C}(e_t) && \{\text{dist. of } \sqcap \text{ over } \sqcup\} \\
&= \left(\mathcal{C}(e_s) \sqcap \bigsqcup_{h \in \Gamma_A(e_s,e_t)} \mathcal{C}(h)\right) \sqcup \mathcal{C}(e_t) && \{\text{defn. of } \mathsf{Act}\} \\
&= \mathsf{Act}\left(\mathcal{C}(e_s), \bigsqcup_{h \in \Gamma_A(e_s,e_t)} \mathcal{C}(h), \mathcal{C}(e_t)\right)
\end{aligned}$$

The proof of (15) is follows as

$$
\begin{aligned}
&\bigsqcap_{h\in\Gamma_D(e_s,e_t)} \mathsf{Deact}(\mathcal{C}(e_s),\mathcal{C}(h),\mathcal{C}(e_t)) \\
&= \mathsf{Deact}(\mathcal{C}(e_s),\mathcal{C}(h_1^D),\mathcal{C}(e_t)) \sqcap \ldots \sqcap \mathsf{Deact}(\mathcal{C}(e_s),\mathcal{C}(h_n^D),\mathcal{C}(e_t)) && \{\text{defn. of } \mathsf{Deact}\} \\
&= (\neg(\mathcal{C}(e_s) \sqcap \mathcal{C}(h_1^D)) \sqcap \mathcal{C}(e_t)) \sqcap \ldots \sqcap (\neg(\mathcal{C}(e_s) \sqcap \mathcal{C}(h_n^D)) \sqcap \mathcal{C}(e_t)) && \{\sqcap \text{ is involutive}\} \\
&= \neg(\mathcal{C}(e_s) \sqcap \mathcal{C}(h_1^D)) \sqcap \ldots \sqcap \neg(\mathcal{C}(e_s) \sqcap \mathcal{C}(h_n^D)) \sqcap \mathcal{C}(e_t) && \{\text{De Morgan law}\} \\
&= (\neg\mathcal{C}(e_s) \sqcup \neg\mathcal{C}(h_1^D)) \sqcap \ldots \sqcap (\neg\mathcal{C}(e_s) \sqcup \neg\mathcal{C}(h_n^D)) \sqcap \mathcal{C}(e_t) && \{\text{dist. of } \sqcup \text{ over } \sqcap\} \\
&= \left(\neg\mathcal{C}(e_s) \sqcup \bigsqcap_{h\in\Gamma_D(e_s,e_t)} \neg\mathcal{C}(h)\right) \sqcap \mathcal{C}(e_t) && \{\text{De Morgan law}\} \\
&= \neg\left(\mathcal{C}(e_s) \sqcap \bigsqcup_{h\in\Gamma_D(e_s,e_t)} \mathcal{C}(h)\right) \sqcap \mathcal{C}(e_t) && \{\text{defn of } \mathsf{Deact}\} \\
&= \mathsf{Deact}\left(\mathcal{C}(e_s), \bigsqcup_{h\in\Gamma_D(e_s,e_t)} \mathcal{C}(h), \mathcal{C}(e_t)\right)
\end{aligned}
$$

□

The following corollary is immediate from the fact that Δ_C is closed under operations $\neg$, $\sqcap$ and $\sqcup$ as shown in Theorem 1. As before, it ensures that if all edges in the system are initially not inconsistent, then their combination does not produce inconsistent values.

Corollary 1. *Let M be any (A,Σ)-PRG. If M is totally consistent, then, for $e_s, e_t \in E$ and $\odot \in \{\sqcap, \sqcup\}$ it follows that* $\mathsf{gather}^A_{\odot}$, $\mathsf{gather}^D_{\odot}(e_1,e_2) \in \Delta_C$.

Proof. The proof follows from Theorem 1 and the definition of $\mathsf{gather}^A_{\odot}$ and $\mathsf{gather}^D_{\odot}$. □

The next definition introduces functions $\mathsf{gather}^A_{\odot_a,\odot_d}$ and $\mathsf{gather}^D_{\odot_a,\odot_d}$, which follow the common conflict resolution strategy of giving precedence to activating and deactivating hyper-edges, respectively.

Definition 17. *Let M be an (A,Σ)-PRG and $\odot_a, \odot_d \in \{\sqcap, \sqcup\}$. The function*

$$
\begin{aligned}
\mathsf{gather}^A_{\odot_a,\odot_d}(e_s,e_t) \;:\; E \times E &\longrightarrow A^2 \\
(e_s,e_t) &\longmapsto \begin{cases} \mathsf{gather}^A_{\odot_a}(e_s,e_t) & \text{if } \Gamma_A(e_s,e_t) \neq \emptyset \\ \mathsf{gather}^D_{\odot_d}(e_s,e_t) & \text{otherwise} \end{cases}
\end{aligned}
$$

resolves conflicting hyper-edges by prioritizing activating hyper-edges when present. Similarly, the function

$$\mathsf{gather}^{D}_{\odot_a,\odot_d}(e_s,e_t) : E \times E \longrightarrow A^2$$
$$(e_s,e_t) \longmapsto \begin{cases} \mathsf{gather}^{D}_{\odot_a}(e_s,e_t) & \textit{if } \Gamma_D(e_s,e_t) \neq \emptyset \\ \mathsf{gather}^{A}_{\odot_d}(e_s,e_t) & \textit{otherwise} \end{cases}$$

resolves conflicts by prioritizing deactivating hyper-edges when present.

Intuitively, function $\mathsf{gather}^{A}_{\odot_a,\odot_d}(e_s,e_t)$ considers only the activating hyper-edges if any are present (i.e., $\Gamma_A(e_s,e_t) \neq \emptyset$), and $\mathsf{gather}^{D}_{\odot_a,\odot_d}(e_s,e_t)$ follows the opposite strategy, giving priority to deactivating hyper-edges.

As stated, we are interested in combining information from all hyper-edges when resolving conflicts. Hence, we introduce the function $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}$, which, unlike the functions in Definition 17, does not prioritize one type of hyper-edges over the other. Thus, whenever there is a conflict between hyper-edges of different types, as illustrated in Figure 5, the function $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}$ first combines the information from activating and deactivating hyper-edges separately, and then combines both results.

Definition 18. *Let M be an (A,Σ)-PRG and $\odot \in \{\sqcap,\sqcup,\otimes,\oplus\}$ and $\odot_a,\odot_d \in \{\sqcap,\sqcup\}$. The function $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}(e_s,e_t) : E \times E \longrightarrow A^2$ combines the information from all conflicting hyper-edges that are triggered by e_s and target e_t as follows*

$$\mathsf{gather}^{\odot}_{\odot_a,\odot_d}(e_s,e_t) = \begin{cases} \mathsf{gather}^{A}_{\odot_a}(e_s,e_t) & \textit{if } \Gamma_A(e_s,e_t) \neq \emptyset \textit{ and } \Gamma_D(e_s,e_t) = \emptyset \\ \mathsf{gather}^{D}_{\odot_d}(e_s,e_t) & \textit{if } \Gamma_A(e_s,e_t) = \emptyset \textit{ and } \Gamma_D(e_s,e_t) \neq \emptyset \\ \mathsf{gather}^{A}_{\odot_a}(e_s,e_t) \odot \mathsf{gather}^{D}_{\odot_d}(e_s,e_t) & \textit{otherwise} \end{cases}$$

Informally, if there are only activating hyper-edges, we combine their information using the function $\mathsf{gather}^{A}_{\odot_a}$. Similarly, if there are deactivating hyper-edges only, we use $\mathsf{gather}^{D}_{\odot_d}$. Finally, if there is a mixture of activating and deactivating hyper-edges in $\Gamma(e_s,e_t)$, we first combine each type separately using $\mathsf{gather}^{A}_{\odot_a}$ and $\mathsf{gather}^{D}_{\odot_d}$ respectively, and then combine the results using $\odot$.

This allows us to have sixteen ways to combine conflicts with function $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}$ since $\odot_a,\odot_d \in \{\sqcap,\sqcup\}$ and $\odot \in \{\sqcap,\sqcup,\otimes,\oplus\}$. The different choices give rise to different policies that should align with the problem at hand.

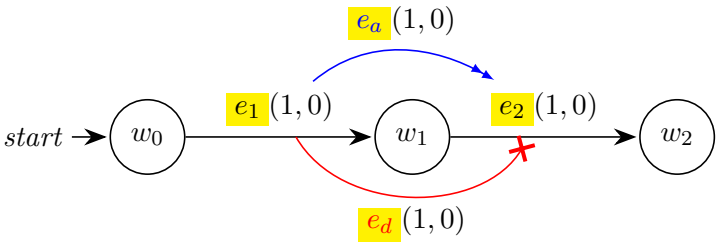
The choice of operators $\odot_a$ and $\odot_d$ has been addressed before and it boils down to whether we value any available evidence or require consensus among hyper-edges for the activation or deactivation of an edge. Whereas the choice of $\odot$, which is relevant when $\Gamma_A(e_s,e_t) \neq \emptyset$ and $\Gamma_D(e_s,e_t) \neq \emptyset$, essentially carries the same intuition and meaning as the operator $\odot$ in the twist-structure.

That is, $\mathsf{gather}^{\sqcap}_{\odot_a,\odot_d}$ (resp. $\mathsf{gather}^{\sqcup}_{\odot_a,\odot_d}$) corresponds to the usual "and" (resp. "or"), so the function performs the conjunction (resp. disjunction) of the information given by the activating hyper-edges and by the deactivating hyper-edges, which corresponds again to the "worst-case" (resp. "best-case") scenario.

Function $\mathsf{gather}^{\otimes}_{\odot_a,\odot_d}$ boils down to a "consensus" between the information given by the activating and deactivating hyper-edges. Meanwhile, $\mathsf{gather}^{\oplus}_{\odot_a,\odot_d}$ stands for "accept all", i.e., it combines all the knowledge provided by the activating and deactivating hyper-edges.

The following example illustrates how information from conflicting hyper-edges is combined using the function $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}$.

Example 16. *Consider the following* $(\{0,1\},\emptyset)$-PRG[6].



This is a scenario involving conflicting hyper-edges: according to e_a*, the target edge* e_2 *should be fully activated, while according to* e_d*, it should be fully deactivated. Formally,*

$$\mathsf{gather}^{A}_{\odot_a}(e_1,e_2) \;=\; \mathsf{Act}(\mathcal{C}(e_1),\mathcal{C}(e_a),\mathcal{C}(e_2)) \;=\; (1,0)$$
$$\mathsf{gather}^{D}_{\odot_d}(e_1,e_2) \;=\; \mathsf{Deact}(\mathcal{C}(e_1),\mathcal{C}(e_d),\mathcal{C}(e_2)) \;=\; (0,1)$$

Let us see the result of combining the information from e_a *and* e_d *using the function* $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}(e_s,e_t)$*, while varying the choice of operator* $\odot$*. Since the conflict involves only one activating and one deactivating hyper-edge, the specific choices of* $\odot_a$ *and* $\odot_d$ *are irrelevant.*

Therefore, following the traversal of e_1*, the value of* e_2 *is updated according to the choice of* $\odot$*. This results in the following possibilities:*

- $\mathsf{gather}^{\sqcap}_{\odot_a,\odot_d}(e_s,e_t) = (1,0) \sqcap (0,1) = (0,1)$ *which yields the most conservative or "worst-case" scenario where* e_2 *is completely deactivated.*

[6]Note that since all transitions are classically true, i.e. $(1,0)$, this PRG could be represented as a standard reactive graph.

- $\mathsf{gather}^{\sqcup\!\!\sqcup}_{\odot_a,\odot_d}(e_s,e_t) = (1,0) \sqcup\!\!\sqcup (0,1) = (1,0)$ *which yields the most optimistic or "best-case" scenario where* e_2 *is completely activated.*
- $\mathsf{gather}^{\otimes}_{\odot_a,\odot_d}(e_s,e_t) = (1,0) \otimes (0,1) = (1 \sqcap 0,\ 0 \sqcap 1) = (0,0)$ *represents the "consensus" between all conflicting hyper-edges. Since* e_a *and* e_d *disagree completely, the result reflects complete vagueness/ignorance.*
- $\mathsf{gather}^{\oplus}_{\odot_a,\odot_d}(e_s,e_t) = (1,0) \oplus (0,1) = (1 \sqcup 0,\ 0 \sqcup 1) = (1,1)$ *represents "accepting all" information given by the conflicting hyper-edges, which yields a complete contradiction, the system accepts both the activation and deactivation.*

The following corollary follows from Corollary 1 and Theorems 1 and 2.

Corollary 2. *Let* M *be any* (A,Σ)-PRG *and* $\odot_a, \odot_d \in \{\sqcup\!\!\sqcup, \sqcap\!\!\sqcap\}$.

- *If* M *is totally consistent, then for* $e_s, e_t \in E$ *and* $\odot \in \{\sqcap\!\!\sqcap, \sqcup\!\!\sqcup, \otimes\}$ *it follows that* $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}(e_1,e_2) \in \Delta_C$.
- *If* M *is totally inconsistent, then for* $e_s, e_t \in E$ *and for* $\odot \in \{\sqcap\!\!\sqcap, \sqcup\!\!\sqcup, \oplus\}$ *it follows that* $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}(e_1,e_2) \in \Delta_P$.

Note that the operator $\oplus$ does not preserve consistency, and $\otimes$ does not preserve knowledge. Hence, choosing to "accept all" information may result in inconsistencies, while opting for "gullibility" may lead to a loss in the overall amount of knowledge.

We now aim to define a semantics by "unfolding" any PRG into a paraconsistent graph. As before, we first describe how the activation function $\mathcal{C}$ of a PRG is updated after traversing an edge, using the conflict resolution functions introduced in Definitions 17 and 18.

Definition 19. *Let* M *be an* (A,Σ)-PRG*, and let* $\mathcal{C}$ *its current activation function. Let* $\odot_a, \odot_d \in \{\sqcap\!\!\sqcap, \sqcup\!\!\sqcup\}$ *and* $\odot \in \{\sqcap\!\!\sqcap, \sqcup\!\!\sqcup, \otimes, \oplus\}$ *be operators defining the conflict resolution policies. For any edges* $e, x \in E$*, we define the updated value of edge* x *after traversing edge* e *as follows:*

$$\mathcal{C}^{e}_{A,\odot_a,\odot_d}(x) = \begin{cases} \mathcal{C}(x) & \text{if } x \notin (\mathsf{on}(e) \cap \mathsf{off}(e)) \\ \mathsf{gather}^{A}_{\odot_a,\odot_d}(e,x) & \text{otherwise} \end{cases} \tag{17}$$

when conflicts are resolved using the function $\mathsf{gather}^{A}_{\odot_a,\odot_d}$.

$$\mathcal{C}^{e}_{D,\odot_a,\odot_d}(x) = \begin{cases} \mathcal{C}(x) & \text{if } x \notin (\mathsf{on}(e) \cap \mathsf{off}(e)) \\ \mathsf{gather}^{D}_{\odot_a,\odot_d}(e,x) & \text{otherwise} \end{cases} \tag{18}$$

when conflicts are resolved using the function $\mathsf{gather}^{D}_{\odot_a,\odot_d}$.

$$\mathcal{C}^{e}_{\odot,\odot_a,\odot_d}(x) \;=\; \begin{cases} \mathcal{C}(x) & \text{if } x \notin (\mathsf{on}(e) \cap \mathsf{off}(e)) \\ \mathsf{gather}^{\odot}_{\odot_a,\odot_d}(e,x) & \text{otherwise} \end{cases} \tag{19}$$

when conflicts are resolved using the function $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}$.

In other words, if edge x is not targeted by any hyper-edge triggered by e, its value remains unchanged. Otherwise, the information from all hyper-edges triggered by e and targeting x is combined using one of the functions: $\mathsf{gather}^{A}_{\odot_a,\odot_d}$, $\mathsf{gather}^{D}_{\odot_a,\odot_d}$, or $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}$, depending on the chosen policies.

We are now equipped to define the semantics for PRG with possibly conflicting hyper-edges. As before, we begin from an initial configuration $\langle w_0, \mathcal{C}_0\rangle$. Then, for any ground edge $e \in \twoheadrightarrow$ originating from the initial state w_0, we compute the updated activation function for all edges x using the update functions defined in Definition 19, according to the chosen conflict resolution policy.

Definition 20. *The semantics of an* (A, Σ)*-paraconsistent graph* P *is defined as the evolution of a configuration* $\langle w, \mathcal{C}\rangle$*, where* $w \in W$ *and* $\mathcal{C} : E \to A^2$*, starting from* $\langle w_0, \mathcal{C}_0\rangle$*. Depending on the conflict resolution policy employed, the system evolves according to the following rule:*

- *If using* $\mathsf{gather}^{A}_{\odot_a,\odot_d}$*:*

$$\frac{\exists e \in \twoheadrightarrow \;\cdot\; \overline{e} = (w, a, w')}{\langle w, \mathcal{C}\rangle \xrightarrow{a, \mathcal{C}(e)}_P \langle w', \mathcal{C}^{e}_{A,\odot_a,\odot_d}\rangle}$$

- *If using* $\mathsf{gather}^{D}_{\odot_a,\odot_d}$*:*

$$\frac{\exists e \in \twoheadrightarrow \;\cdot\; \overline{e} = (w, a, w')}{\langle w, \mathcal{C}\rangle \xrightarrow{a, \mathcal{C}(e)}_P \langle w', \mathcal{C}^{e}_{D,\odot_a,\odot_d}\rangle}$$

- *If using* $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}$*:*

$$\frac{\exists e \in \twoheadrightarrow \;\cdot\; \overline{e} = (w, a, w')}{\langle w, \mathcal{C}\rangle \xrightarrow{a, \mathcal{C}(e)}_P \langle w', \mathcal{C}^{e}_{\odot,\odot_a,\odot_d}\rangle}$$

where $\odot_a, \odot_d \in \{\sqcup, \sqcap\}$ *and* $\odot \in \{\sqcap, \sqcup, \otimes, \oplus\}$.

If the paraconsistent reactive graph does not contain conflicting hyper-edges, then the conflict resolution policies introduced through the functions $\mathsf{gather}^{A}_{\odot_a,\odot_d}$, $\mathsf{gather}^{D}_{\odot_a,\odot_d}$, and $\mathsf{gather}^{\odot}_{\odot_a,\odot_d}$ become unnecessary. Consequently, all of the rules in Definition 20 reduce to the semantics for PRG without conflicting hyper-edges, as given in Definition 15, as expected.

Furthermore, the following result is immediate from Corollary 2.

Corollary 3. *Let M be an (A, Σ)-PRG.*

- *If M is totally consistent, then the system evolves without ever generating inconsistency under any of the rules in Definition 20, except when using the resolution function* $\mathsf{gather}^{\oplus}_{\odot_a,\odot_d}$.
- *If M is totally inconsistent, then the system evolves without ever becoming consistent under any of the rules in Definition 20, except when using the resolution function* $\mathsf{gather}^{\otimes}_{\odot_a,\odot_d}$.

Intuitively, the above corollary clarifies that some conflict-resolution policies preserve inconsistency while others preserve consistency. As shown in Theorem 1, operator $\oplus$ does not preserve consistency, whereas Theorem 2 shows that $\otimes$ does not preserve inconsistencies. Consequently, when these operators are used to resolve conflicts between hyper-edges, via $\mathsf{gather}^{\oplus}_{\odot_a,\odot_d}$ and $\mathsf{gather}^{\otimes}_{\odot_a,\odot_d}$, the unfolding rules of Definition 20 proceed in a way that may affect whether the resulting system will remain consistent or inconsistent.

For instance, as illustrated in Example 16, applying $\mathsf{gather}^{\oplus}_{\odot_a,\odot_d}$ (where $\oplus$ is informally described as "accept all") may introduce inconsistencies even when all transitions initially carry consistent values. Likewise, using $\mathsf{gather}^{\otimes}_{\odot_a,\odot_d}$ (with $\otimes$ informally understood as a "consensus" operator) may fail to preserve inconsistent information; that is, transitions that are originally inconsistent may be updated to consistent ones. These behaviours account for the exceptions stated in Corollary 3.

5 Conclusions

This work is part of a broader research agenda on paraconsistent structures [16, 18, 19]. The conference paper it extends [17] first combined this agenda with reactive graphs [27] by introducing reactive graphs with paraconsistent transitions. These allow ground edges to represent consistent, vague, or inconsistent information. The main contribution was the definition of a semantics that "unfolds" an $\mathrm{RG_P}$ into a paraconsistent graph by eliminating hyper-edges while unfolding the system's reactivity.

Introducing paraconsistency into reactive graphs provides a mechanism for modeling evolving systems where information may be incomplete, contradictory, or subject to revision as new evidence emerges. Such scenarios frequently arise in domains involving multiple agents or sources, where beliefs, lies, uncertainty, or disagreement must be reconciled over time.

This paper generalizes reactive graphs with Paraconsistent Transition by introducing paraconsistent reactive graphs, abbreviated as PRG. In PRG, all edges, both ground and hyper-edges, can represent vague, consistent, or inconsistent information. This generalization is motivated by the observation that, if hyper-edges are interpreted as information provided by different sources, then conflicts and inconsistencies naturally arise. Initially, in Section 4.1, we restricted our focus to PRG without conflicting hyper-edges in order to study the isolated effects of activation and deactivation. In Section 4.2, we addressed however the more general setting where multiple hyper-edges may simultaneously target the same edge, potentially conveying contradictory information.

To manage such conflicts, we proposed a range of resolution strategies, ranging from conservative to optimistic, that allow for flexible combinations of information. In contrast to bivalent switch graphs, where conflict resolution typically defaults to a binary choice (e.g., prioritizing activation over deactivation [30, 42]), our framework accommodates more nuanced policies for handling conflicts. Each resolution strategy is grounded in an intuitive conflict policy that should be chosen according to the problem at hand.

Future work will explore refined conflict resolution policies that go beyond simply combining all conflicting hyper-edges. For instance, one may prioritize hyper-edges based on their degree of consistency or informativeness. A possible approach is to rank hyper-edges by how much information they hold and select the most informative or least inconsistent ones. Another option is to assign trust scores (e.g., values in $[0, 1]$) to hyper-edges, reflecting prior knowledge or confidence in their reliability. These enhancements would require revisiting the definitions of the Act and Deact functions (see Example 12), as well as the gather functions introduced in Section 4.2.

Additional ongoing research includes addressing the challenges posed by various application scenarios of paraconsistent structures, spanning diverse fields from robotics to quantum computation [4].

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